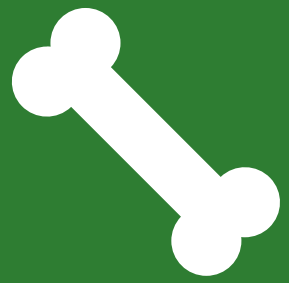
骨科學檢查





森遠動物醫院
SUN YUAN ANIMAL HOSPITAL

個人簡介

林煦倫 獸醫師

現職：森遠犬貓復健中心 待業院長 (裝潢中□)

經歷：小王子動物醫院復健分院 獸醫師

國立臺灣大學附設動物醫院-復健及整合醫學科 獸醫師

寵樂動物醫院 獸醫師

學歷：國立嘉義大學獸醫學系 畢業

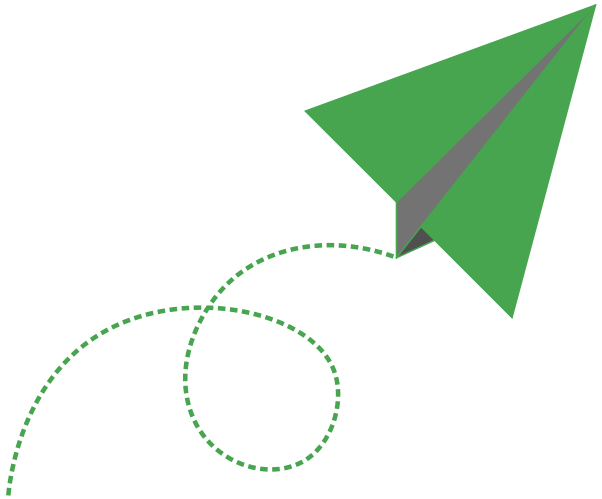
專長：犬貓復健醫學

證照：國際獸醫物理治療師認證 (CCRV)

美國針灸專科獸醫師認證 (CVA)

美國田納西大學犬隻體能訓練師認證 (CCFT)

檢查步驟



1. History and signalment
2. Visual exam
3. Subjective gait analysis
4. Objective gait analysis
5. Palpation

History and signalment

History 病史

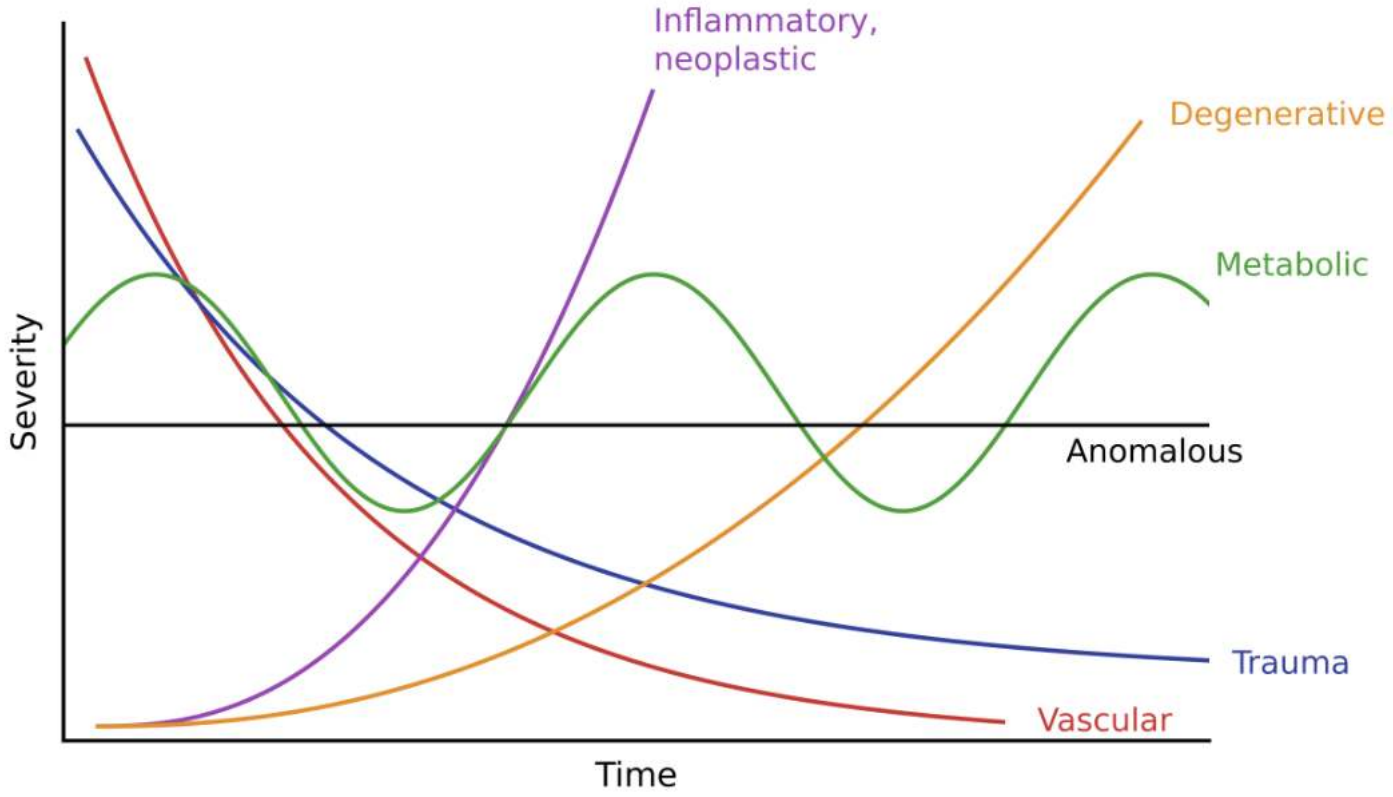
Signalment 重要資訊

History 病史

History 病史

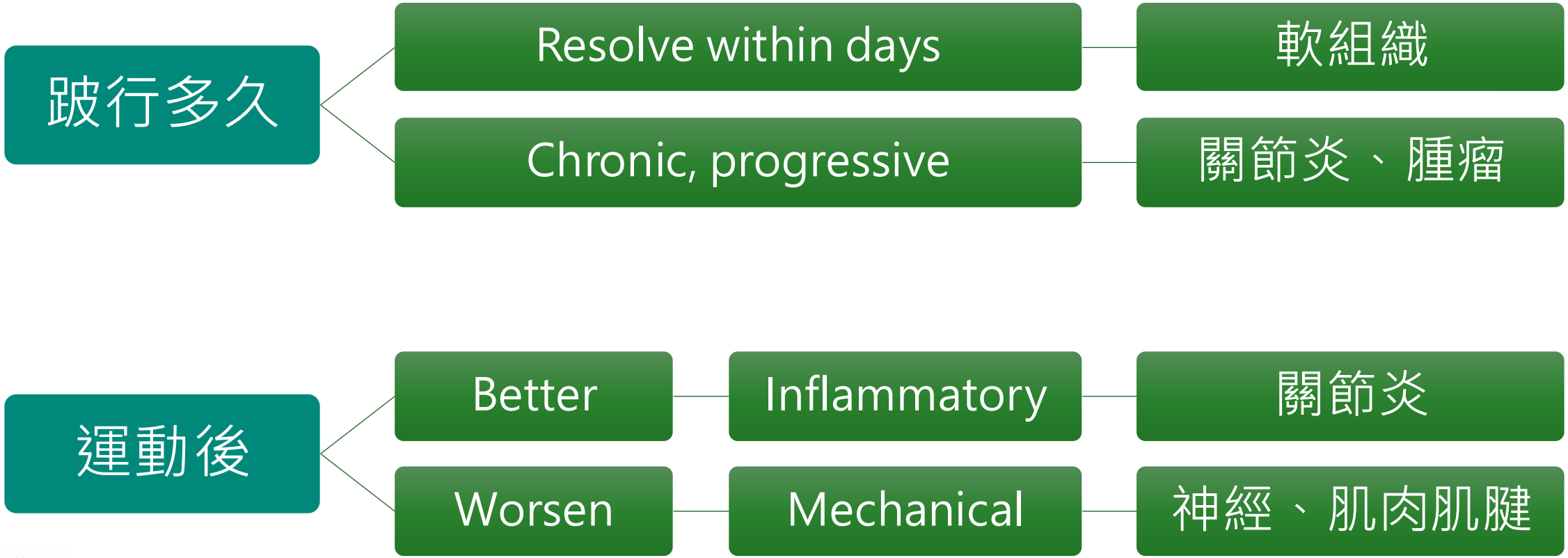
1. Was the inciting cause of the lameness observed ?
2. How long has the lameness been observed ?
3. What was/is your dog's activity level before/since the onset of lameness ?
4. Has the lameness worsened, stayed the same, or improved ?
5. Does the lameness improve or worsen with exercise ?
6. Have any diagnosis been performed ?
7. What treatment(s) has been initiated and what was the response ?

History 病史



Onset and progression of neurologic diseases (adapted from Platt & Olby, 2013)

History 病史


跛行多久
Resolve within days
軟組織
Chronic, progressive
關節炎、腫瘤
運動後
Better
Inflammatory
關節炎
Worsen
Mechanical
神經、肌肉肌腱

Signalment 重要資訊

Signalment 基本資料

- **品種**
  - ✓ 遺傳性疾病
  - ✓ 小型犬 ：膝蓋骨異位
  - ✓ 大型犬 ：髖關節發育不良、骨軟骨病
- **年齡**
  - ✓ 年輕 ：創傷、運動傷害、發育性問題
  - ✓ 老年 ：腫瘤、關節炎
- **性別**
  - ✓ 全骨炎Panosteitis ：雄性犬風險為雌性的四倍
  - ✓ 股骨頭骨骺滑脫症SCFE：絕育公貓較為常見

Visual exam

Visual exam 簡介

Range of motion 關節活動度

Standing posture 站姿

Visual exam 簡介

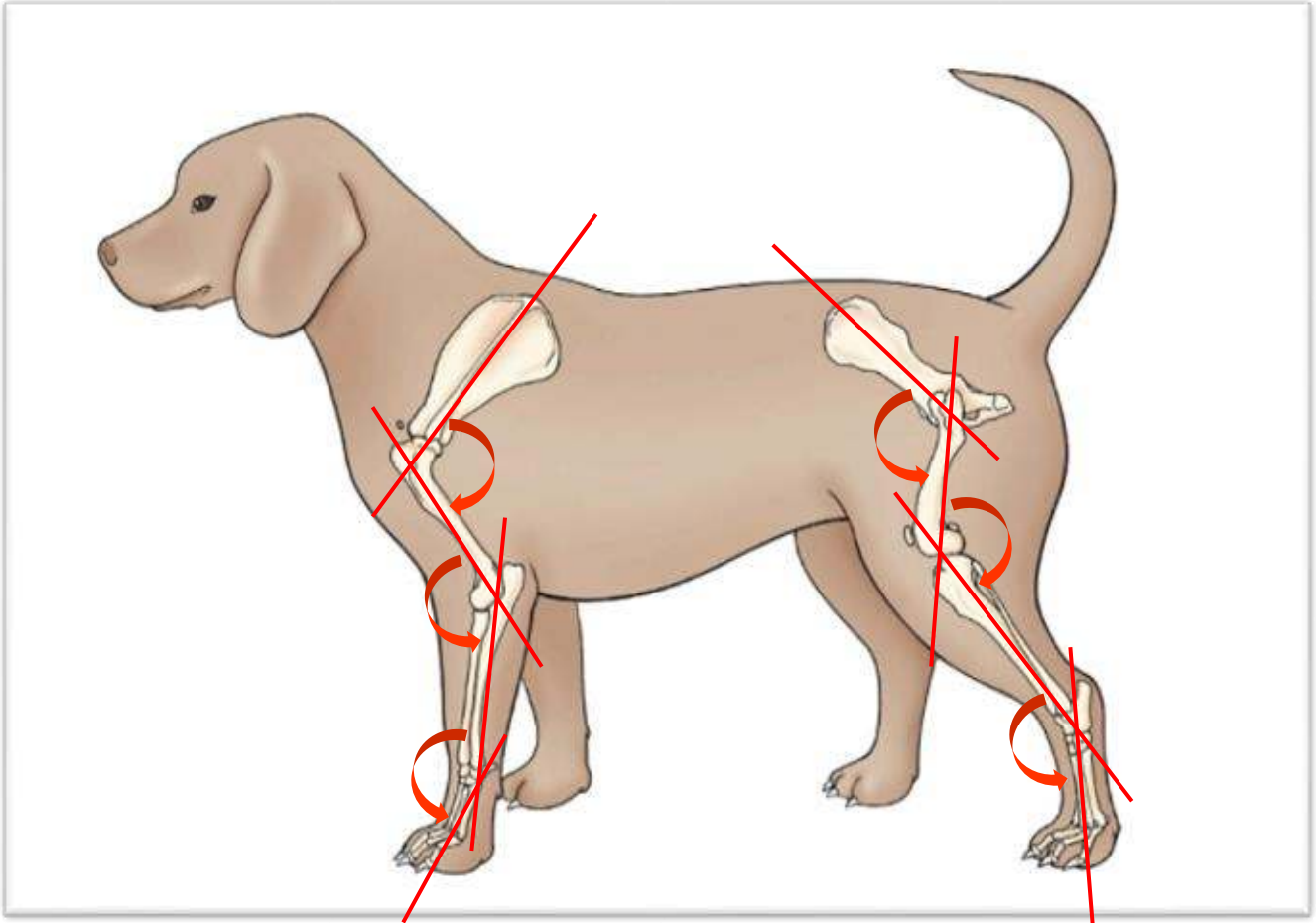
Visual exam

- Off-loading at a stance – affected limb
- Periarticular swelling – joint infusion or chronic degenerative changes
- Hyperextension/Hyperflexion
- Increased joint angle – Achilles tendon rupture

Range of motion 關節活動度

Range of Motion 關節活動度

- 每個關節都有正常的活動範圍，彎曲的最大角度與伸展的最大角度之間，就稱為關節活動度



Range of Motion 關節活動度

關節	伸展Max	彎曲Max
肩關節	165	40
肘關節	165	35
腕關節	195	30
髖關節	160	55
膝關節	160	45
踝關節	170	40



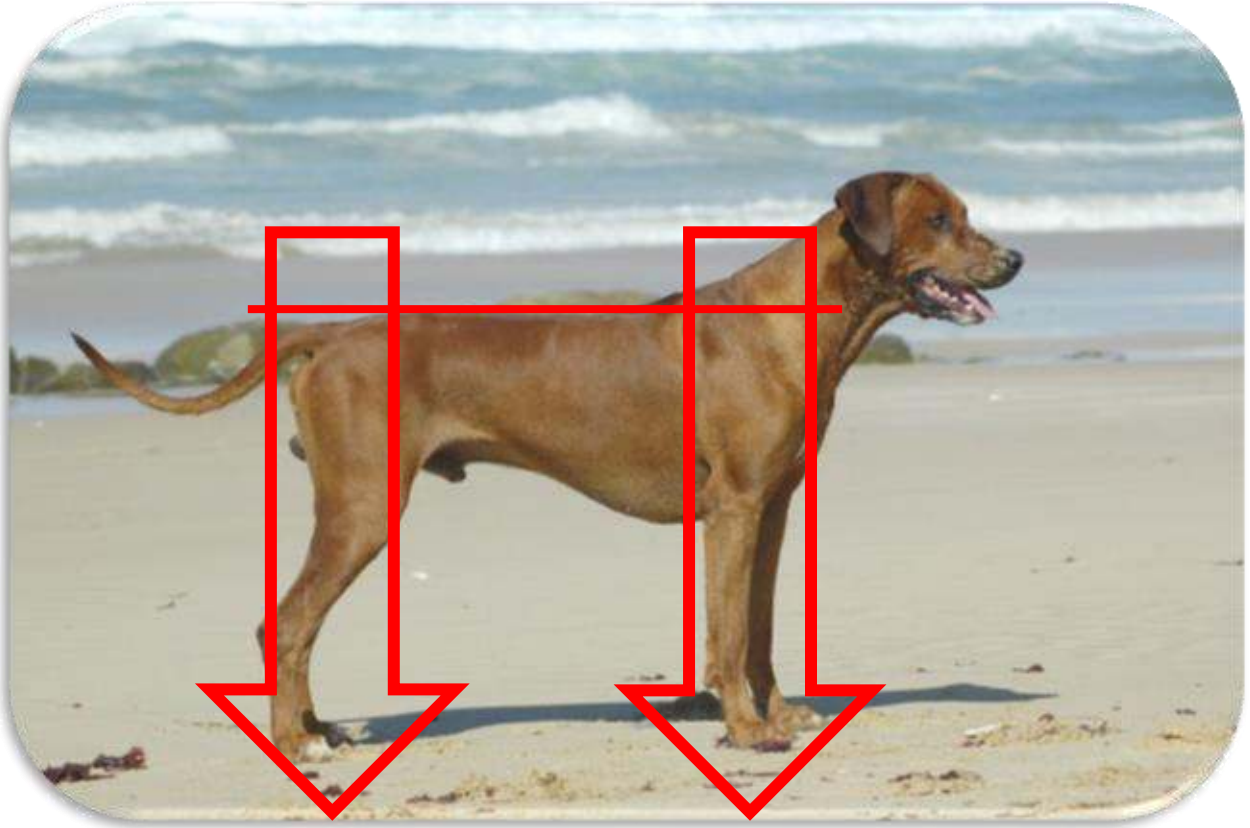
Range of Motion 關節活動度



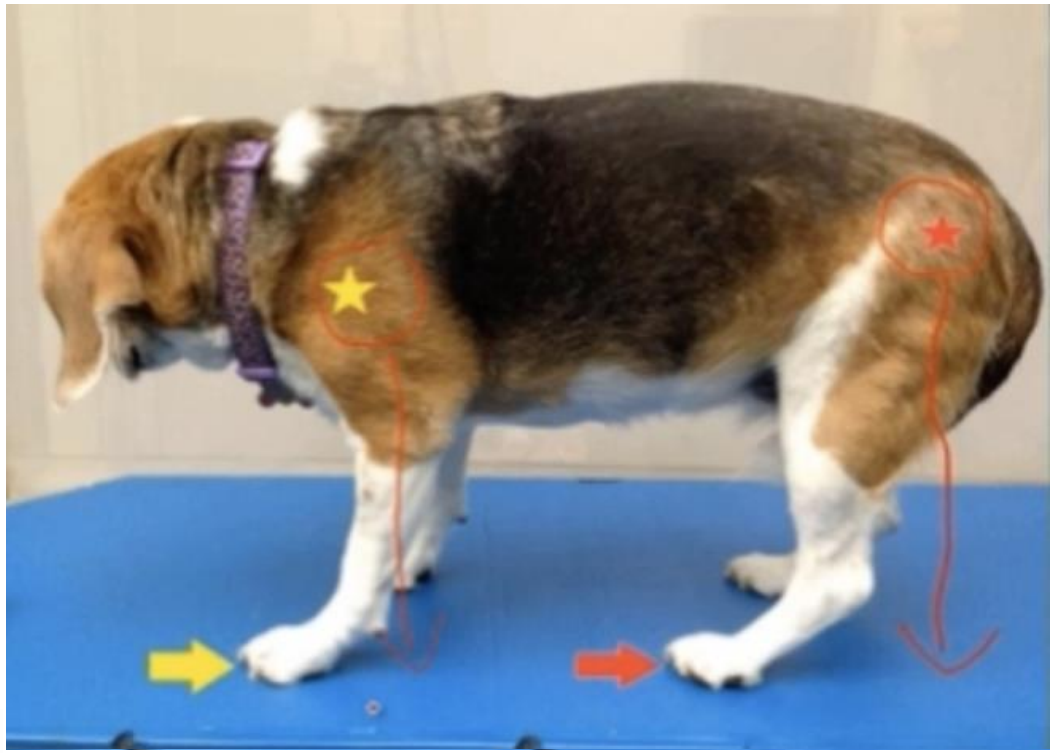
Standing posture 站姿

Standing Posture站姿-正常

- 背部(脊椎)線
- 頭與尾巴
- 骨盆
- 四肢腳
- 負重平衡



Standing Posture站姿-異常



Standing Posture站姿-異常

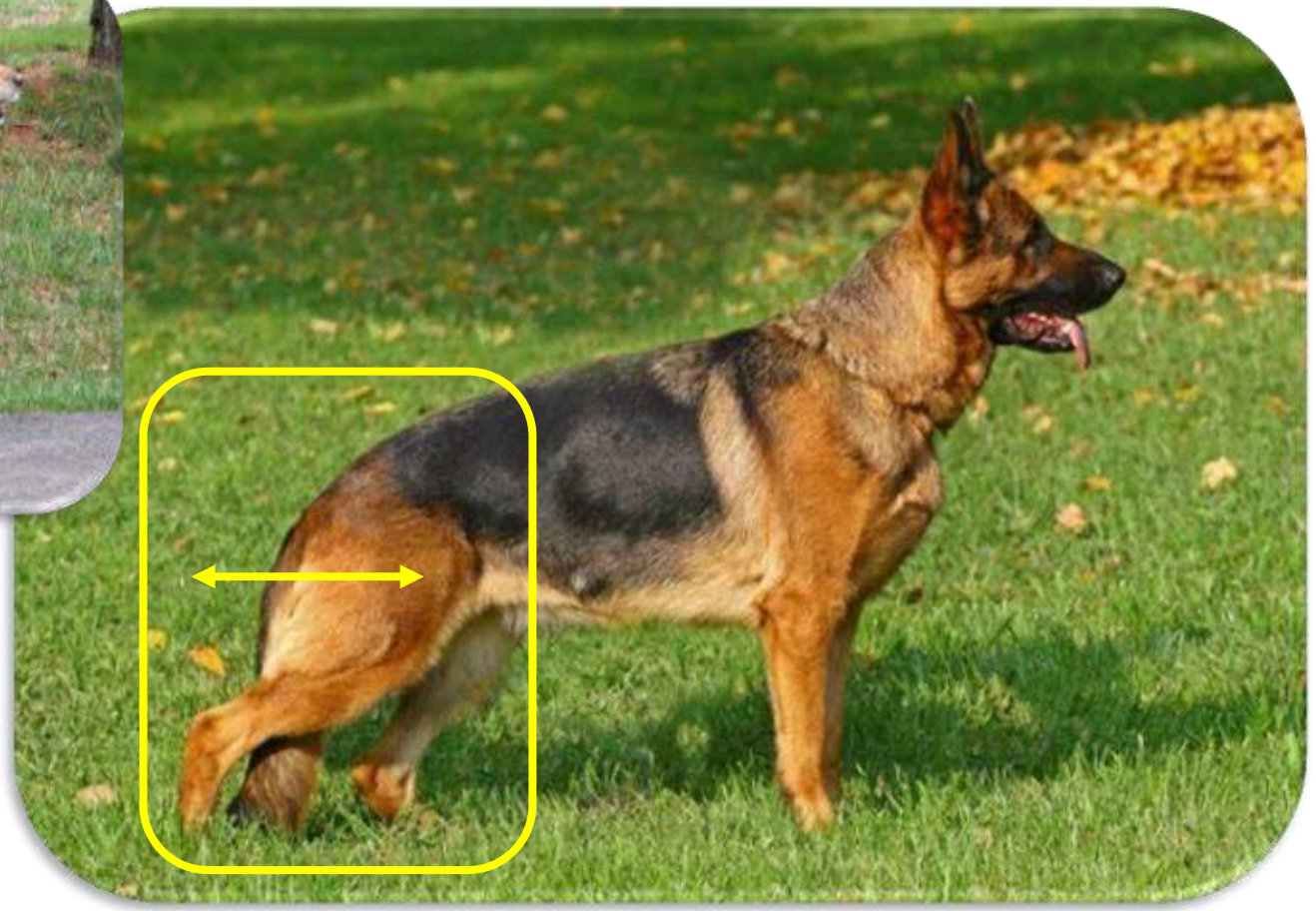


Standing Posture站姿-異常



Standing Posture站姿-正常





Standing Posture站姿-正常



Sitting Posture
坐姿-正常

- 可直接蹲下而坐
- 膝蓋彎曲
- 若不願意坐下，或是腳不願彎曲，有問題!
- Sit-test (for CCLD)



PA8313
DOG
Japanese Spitz

Age : 11 Years
Female

R



Subjective gait analysis

Observation 觀察

Gait pattern 步態

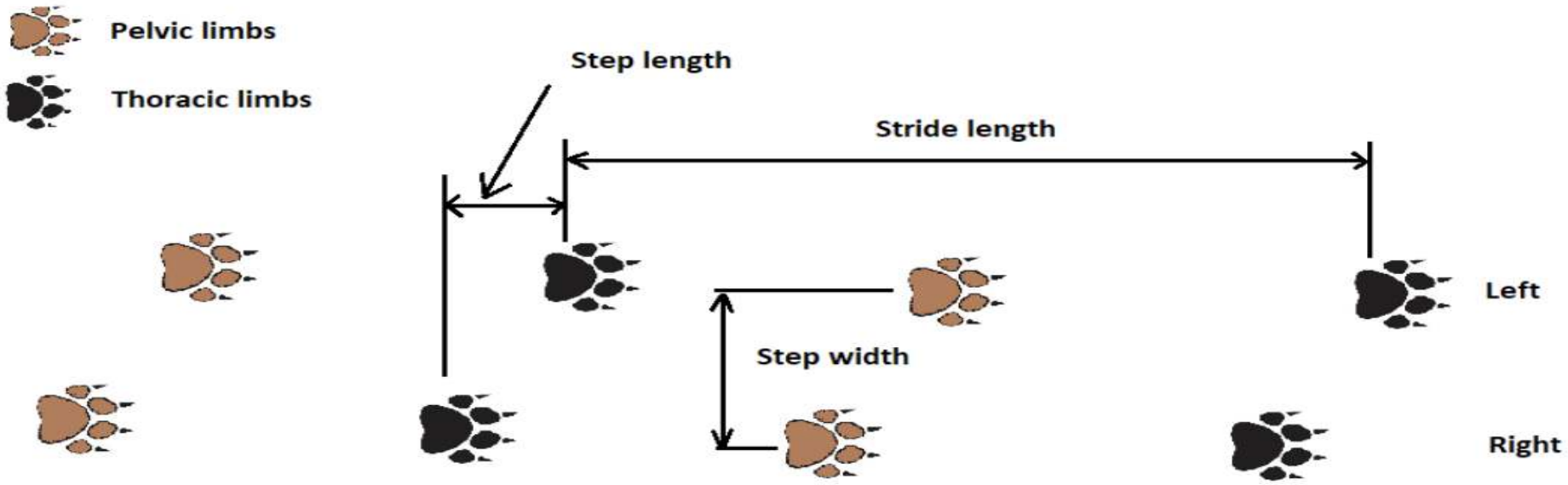
Lameness score 跛行分數

Observation 觀察

Observation - REST

- **Cranial cruciate ligament disease** - Avoiding flexion of the affected stifle(s)
- **Bilateral cruciate disease** - Hesitate to sit down
- **Lumbosacral disease** - Difficulty rising
- **Neurologic disease** - Spontaneous knuckling

Observation - MOTION



- Change in stride/step length – musculoskeletal or neurologic disease

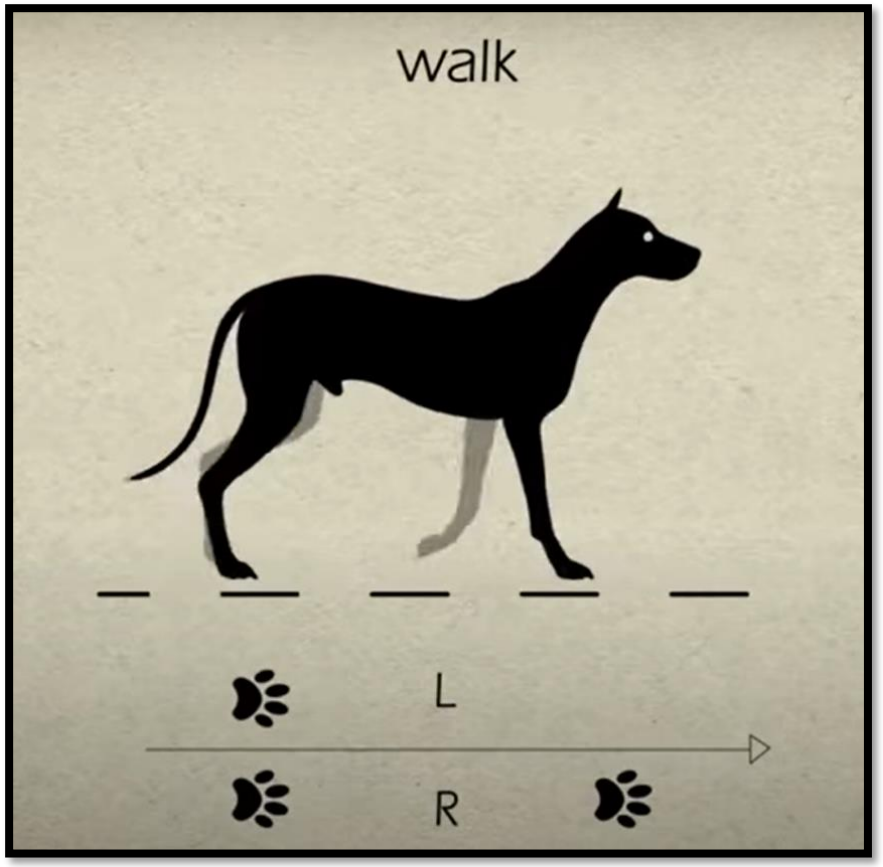
Gait pattern 步態

六種步態的類型

- Walk-走、散步
- Amble-緩行
- Pace-踱步
- Trot-小跑、快步
- Canter-慢跑
- Gallop-奔馳、跑

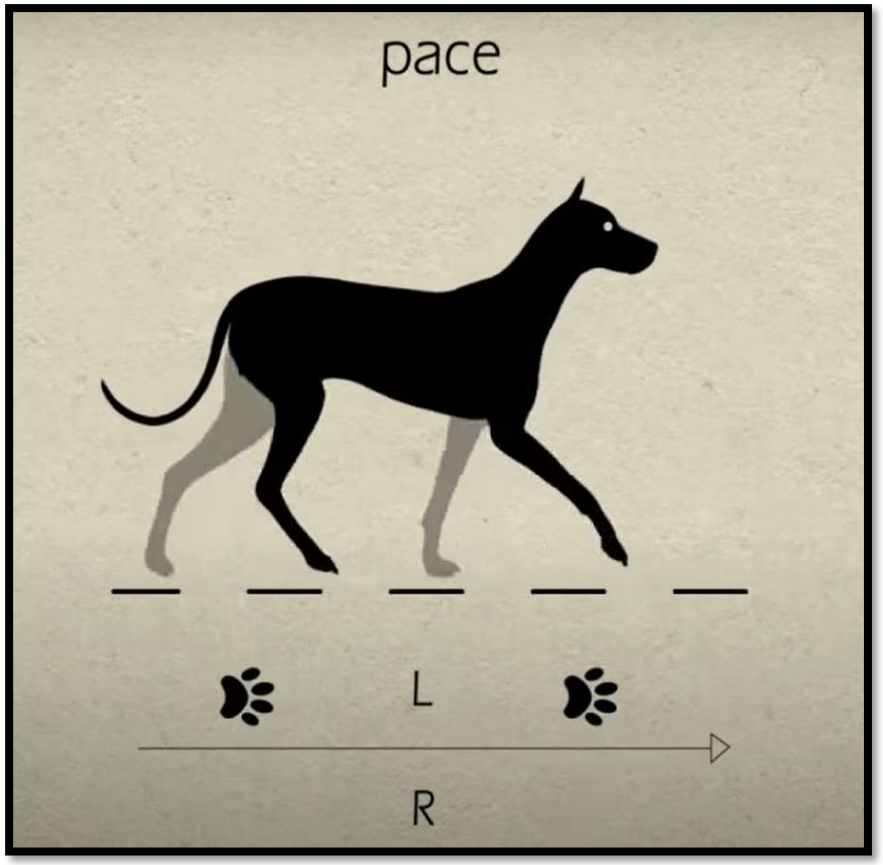
Walk-走、散步

- 速度最慢的步態
- 右後-右前-左後-左前
- 動力來自後腳
- 唯一同時有三隻腳在地上的步態



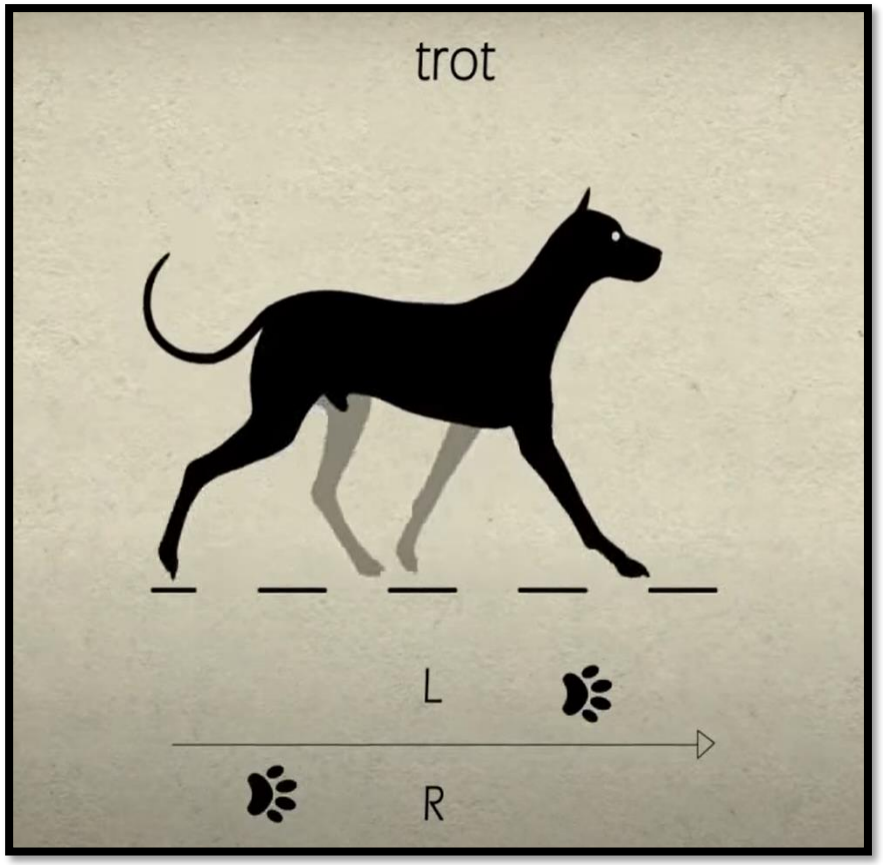
Pace-踱步

- 同側腳一起擺動
- 右後&右前-左後&左前
- 通常為過渡期步態
- 可能為疼痛的代償性步態
- 任何時候只有兩隻腳在地上



Trot-小跑、快步

- 對角線前後腳一起擺動
- 右前&左後-左前&右後
- 最高效率的步態
- 最易觀察輕微負重跛行的步態



轉移重心-辨別患肢小技巧

1. **Vertical head movement**
   - 又叫做 Head nod/bob
   - 通常用來評估”前腳”
   - 頭抬起來那隻腳是疼痛腳
2. **Pelvic movement**
   - 又叫做 Hip hike
   - 通常用來評估”後腳”
   - 屁股抬起來那隻腳是疼痛腳
3. **Tail movement**
   - 尾巴往上甩那隻腳是疼痛腳

Abnormal Gait異常步態-點頭

HIP HIKE & TAIL MOVEMENT

Abnormal Gait異常步態-兔子跳

Lameness score 跛行分數

Table 1.1 Unvalidated numerical rating score used by the author to subjectively quantify canine lameness.

Score	Lameness degree	Lameness description
0	None	*No identifiable lameness* Weight-bearing at all times
1	Slight	*Inconsistent lameness* that is difficult to observe and/or it is difficult to determine the affected limb (i.e. no consistent head movement/pelvic tilt is observed) Weight-bearing at all times
2	Mild	Clearly detectable lameness associated with minor *head movement/pelvic tilt* Weight-bearing at all times
3	Moderate	Clearly detectable lameness associated with obvious *head movement/pelvic tilt* Weight-bearing at all times
4	Severe	Clearly detectable lameness associated with obvious head movement/pelvic tilt *Occasionally non-weight-bearing/toe touching*
5	Non-weight-bearing	*Always non-weight-bearing/toe touching*

Non-weight-bearing

This scoring system can be applied at the walk and/or the trot depending on the patient's clinical status. The patient should only be scored during motion (i.e. off-loading at a stance is not included in this assessment). To increase the sensitivity, the scoring system can be applied for both gaits. If a comparison between different time points is performed, only the scoring within one gait can be compared.

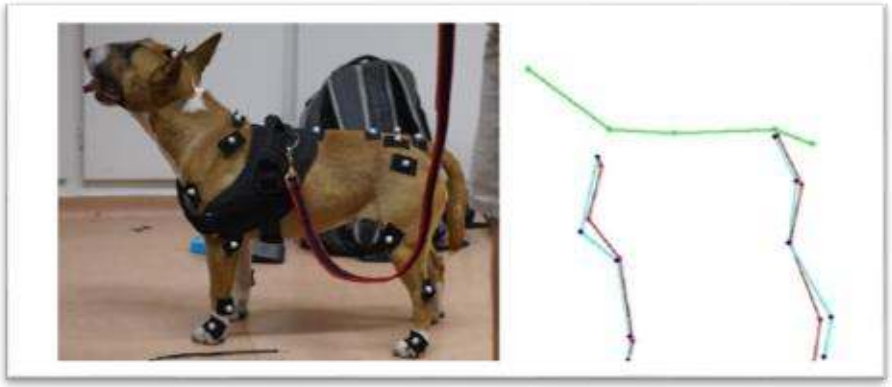
Objective gait analysis

Kinetic 動力學

Kinematic 運動學

Kinetic vs. Kinematic

- 利用客觀的儀器測量步態的各項數據



1. Kinetic – force (力量、壓力)
   - Stance analysis (站立分析)
   - Force plate (測力板)
   - Pressure sensitive walkway (壓力感應走道)



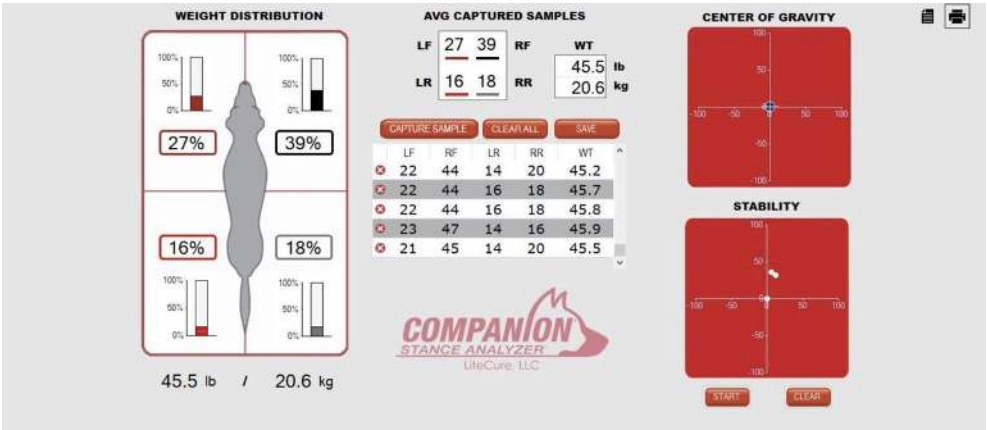
2. Kinematic – motion (運動)
   - Pulsed-LEDs or reflective marker (光點或反光標示)
   - IMU ; Internal measurement unit (慣性測量單元)

Kinetic 動力學

Stance analysis 站立分析

- **優點**
  - ✓ 站立四肢負重力分析
  - ✓ 報告立刻產生
  - ✓ 臨床實用性高
- **缺點**
  - ✓ 僅提供靜態負重百分比%
- **限制性**
  - ✓ 動物安定性及配合度須佳
  - ✓ 小型犬較不適合

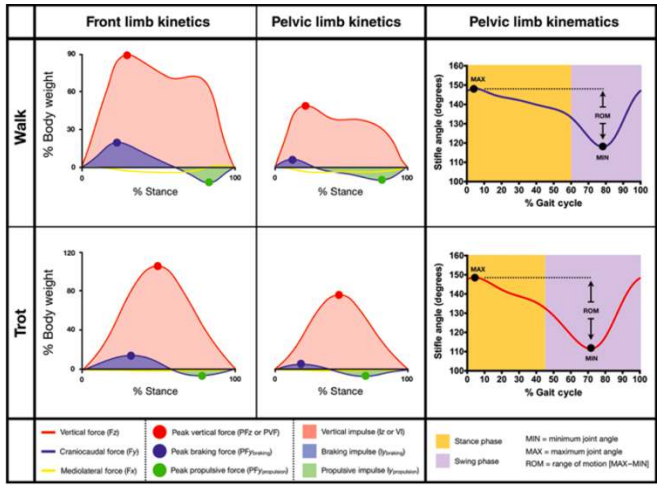


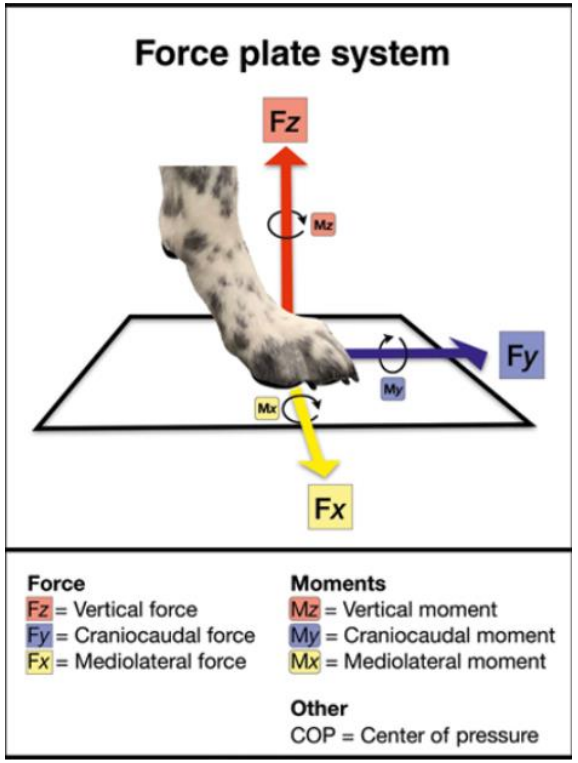


Force plate 測力板

- **優點**
  - ✓ 地面反作用力 (Ground Reaction Force)
  - ✓ **Peak vertical force (PVF)**
  - ✓ 準確性高
  - ✓ X, Y, Z三軸測力
- **缺點**
  - ✓ 臨床實用性低 – 研究用
- **限制性**
  - ✓ 裝置固定不方便移動
  - ✓ 中大型犬較適合

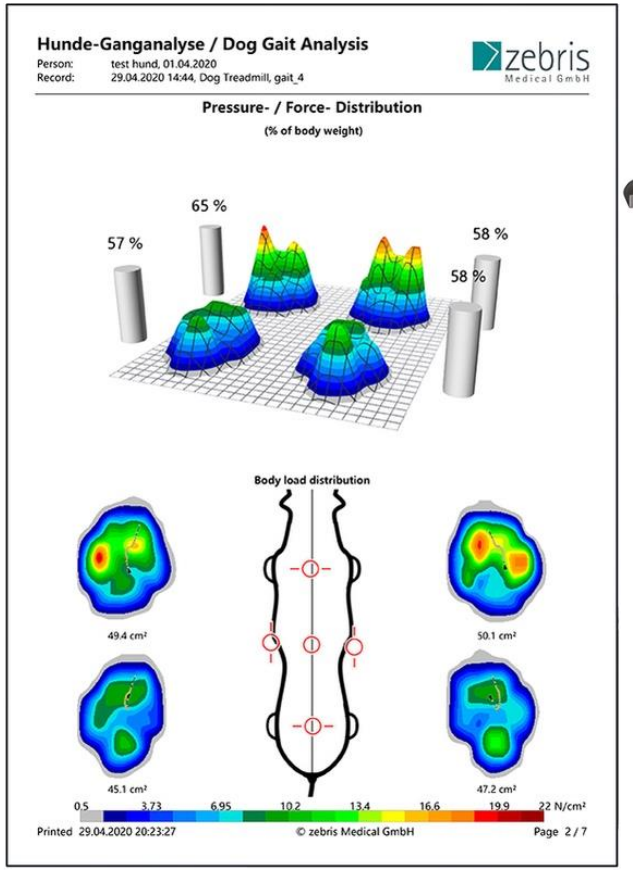




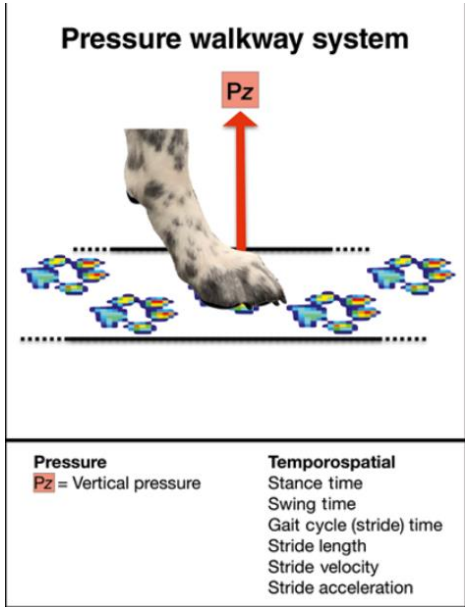


Pressure sensitive walkway 壓力感應走道

- **優點**
  - ✓ 測壓力亦可提供vertical force受力資訊
  - ✓ 報告立刻產生
  - ✓ 臨床實用性高
  - ✓ 小型犬亦適用
- **缺點**
  - ✓ 無法提供關節動態資訊
- **限制性**
  - ✓ 裝置為硬板不適合經常移動



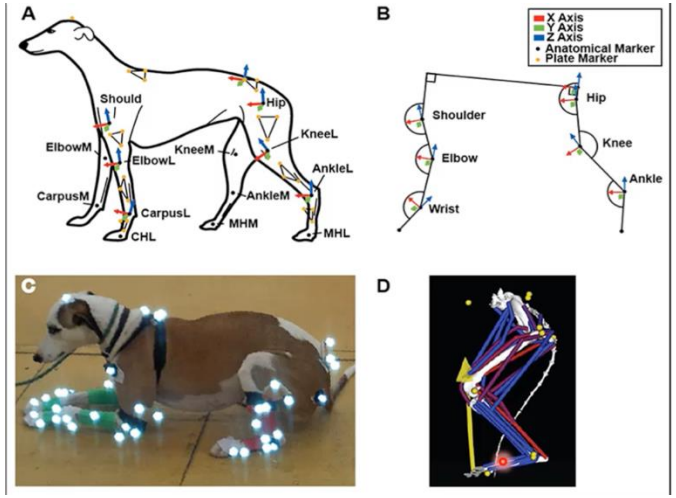


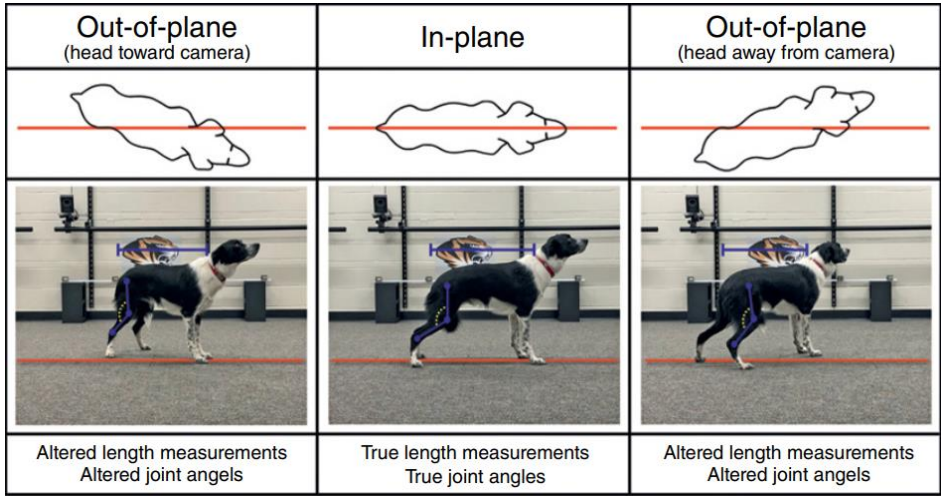


Kinematic 運動學

Kinematic

- **優點**
  - ✓ 關節角度變化
  - ✓ 2D or 3D
- **缺點**
  - ✓ 需剃毛以安置及固定標誌物
  - ✓ 臨床實用性低 – 研究用
- **限制性**
  - ✓ 2D須符合校正線
  - ✓ 3D需多台攝影機
  - ✓ 設備要求度高





Palpation

Ortolani test

Patella luxation

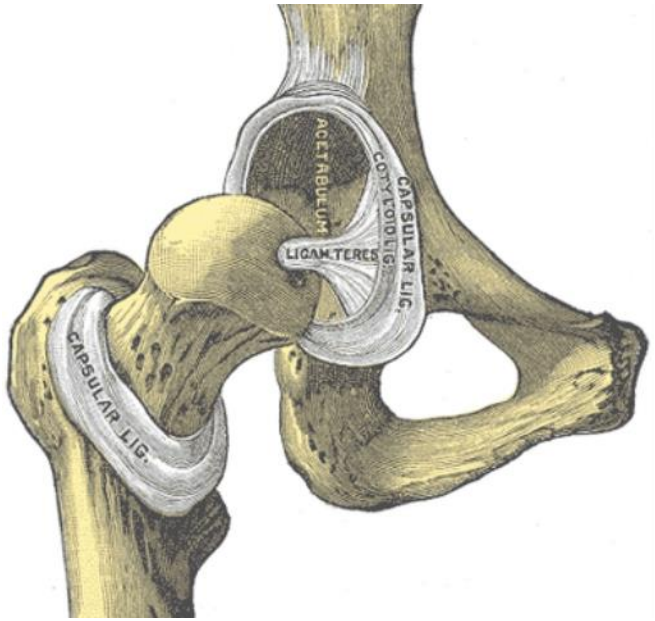
Drawer test & Tibial compression

Varus / Valgus

Biceps tendon

Iliopsoas

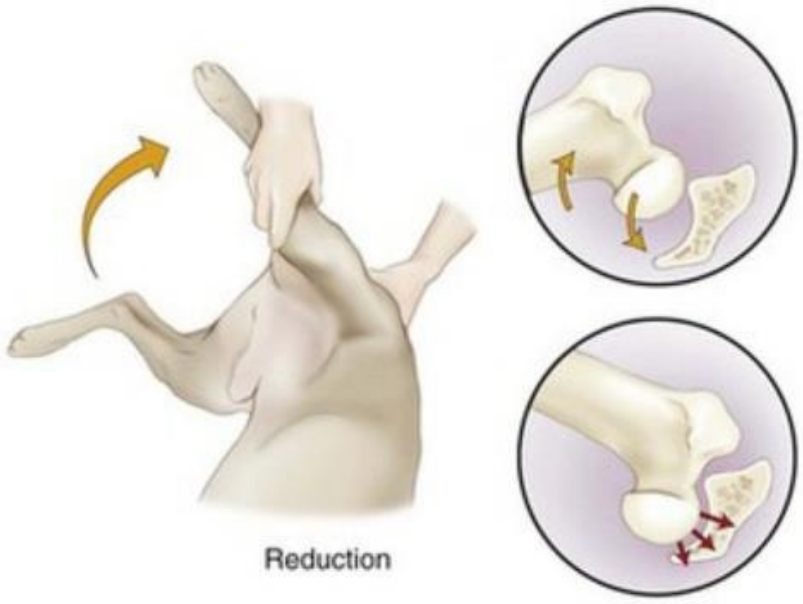
Ortolani test



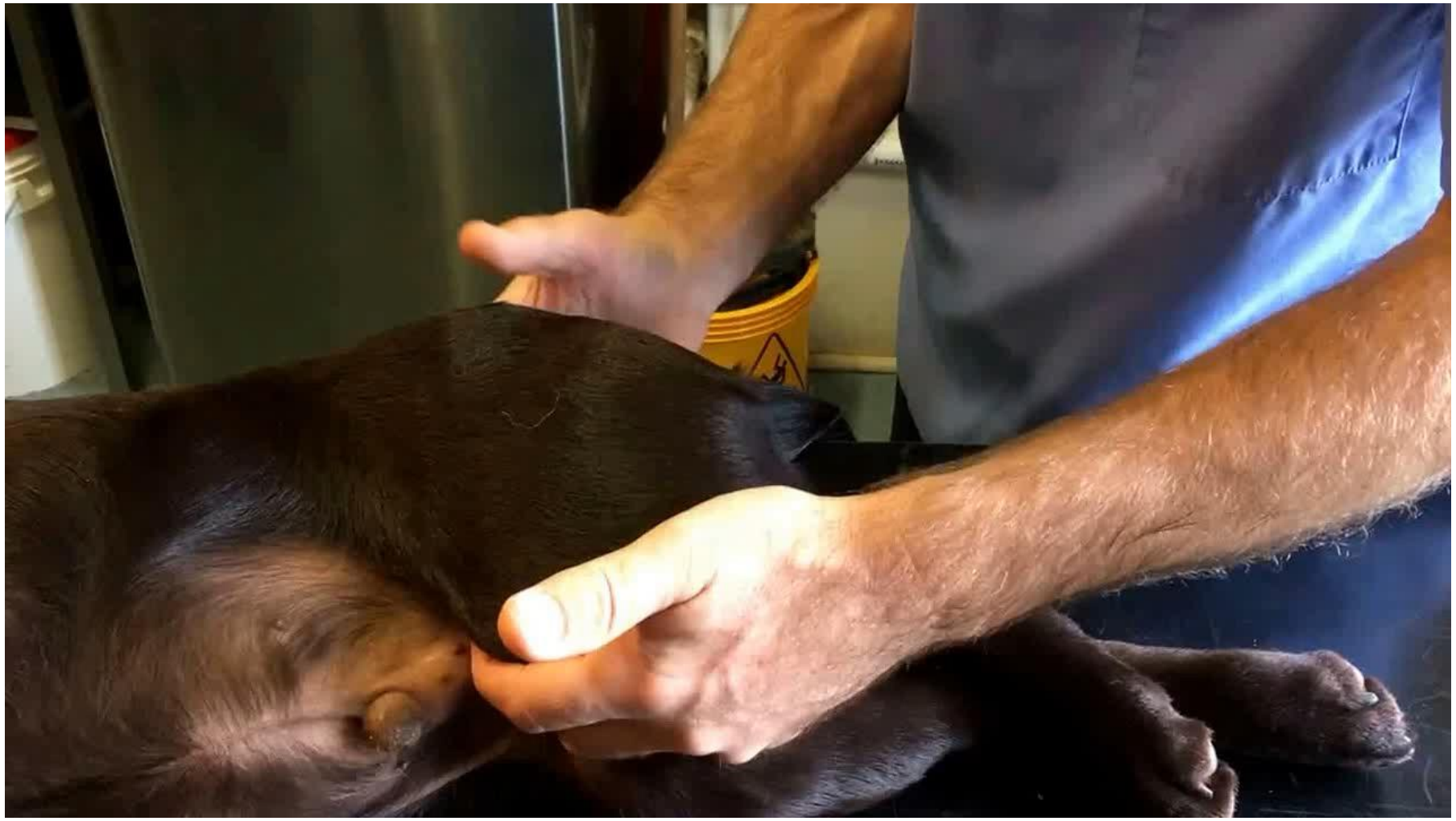
- **通常使犬隻呈現側躺，在使髖關節內展(adduction)時，施加往背側的壓力於髖關節上**

1. 大拇指放在大轉節(Greater trochanter)上
2. 另一隻手抓穩股骨，進行內展的同時將股骨往背側推擠
3. 股骨從髖臼窩脫出，接著進行外展(abduction)
4. 若有半脫位，可感受到喀一聲或大拇指往下陷

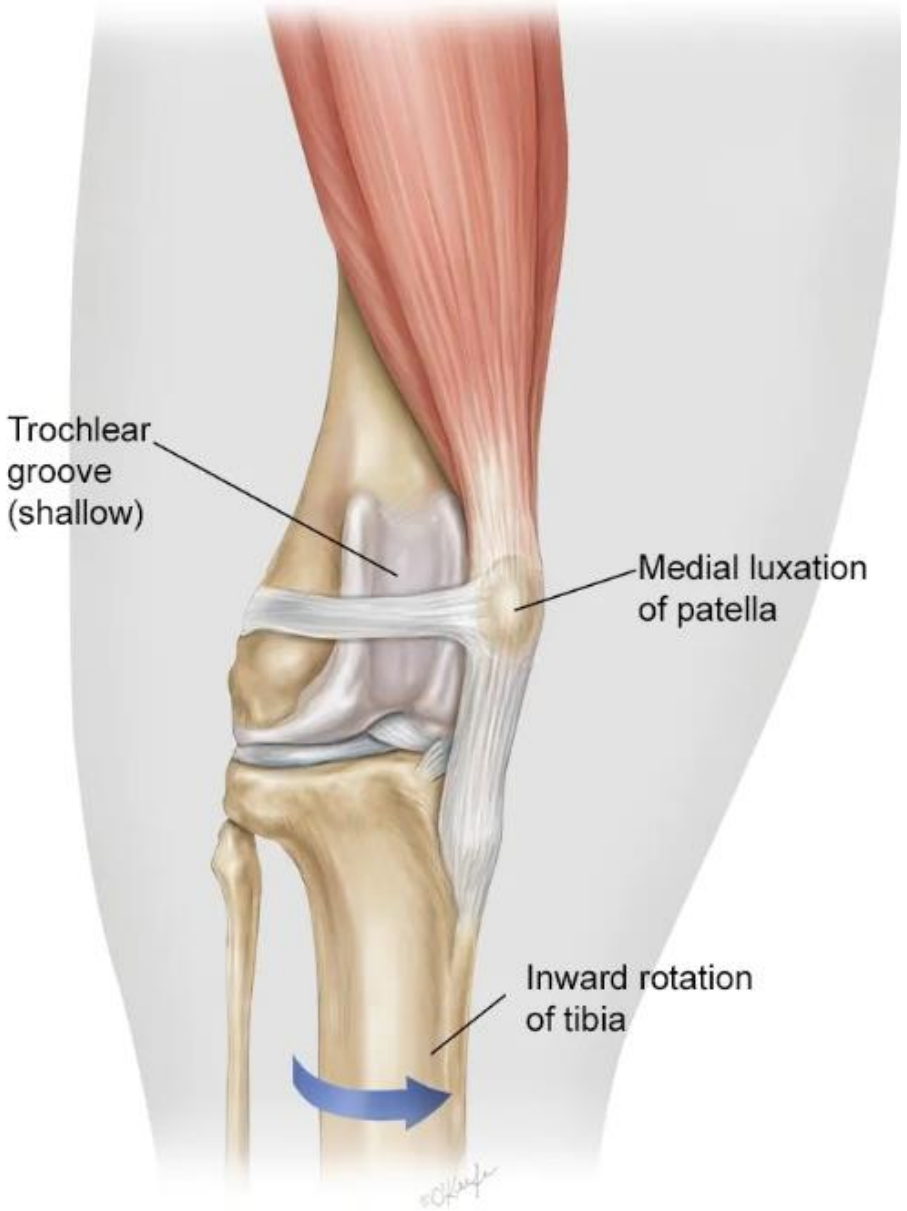


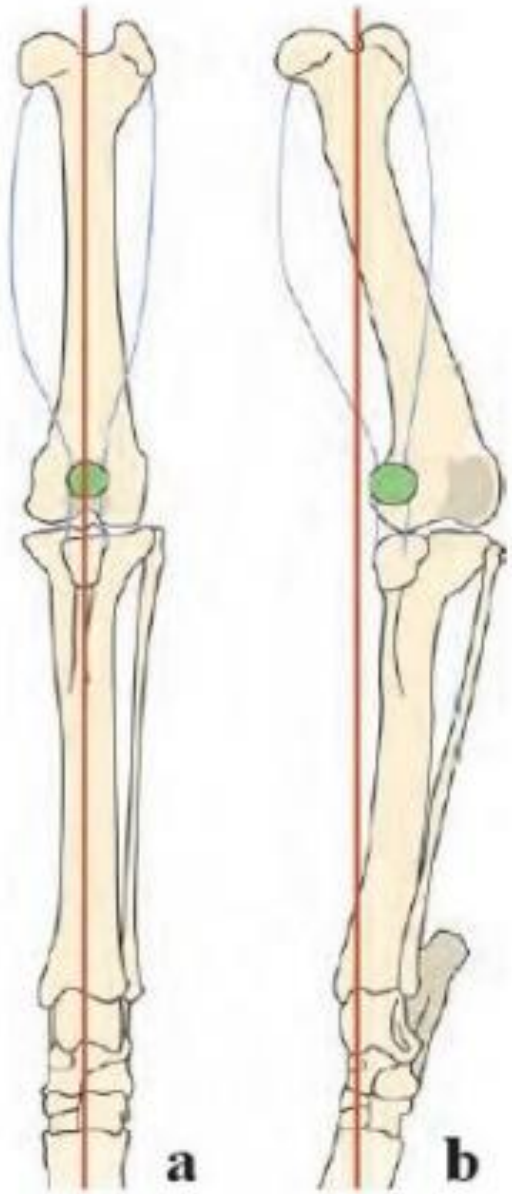


ORTOLANI TEST



Patella luxation





Medial luxation : 伸直+內轉
Lateral luxation : 彎曲+外轉

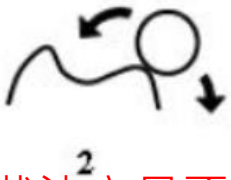
Patella alta (高位髕骨) → MPL
Patella baja (低位髕骨) → LPL

Medial luxation : O型腿
Lateral luxation : 內八

膝蓋骨異位分級



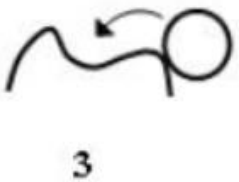
- **第一級**：一般情況下，膝蓋骨在滑車溝內，有外力狀況下，可以被推出滑車溝，外力消失，則回到正常位置



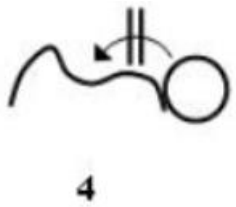
- **第二級**：一般情況下，膝蓋骨在滑車溝內，有外力狀況下，可以被推出滑車溝，外力消失，停留在內側或外側

臨床症狀決定是否手術

手術

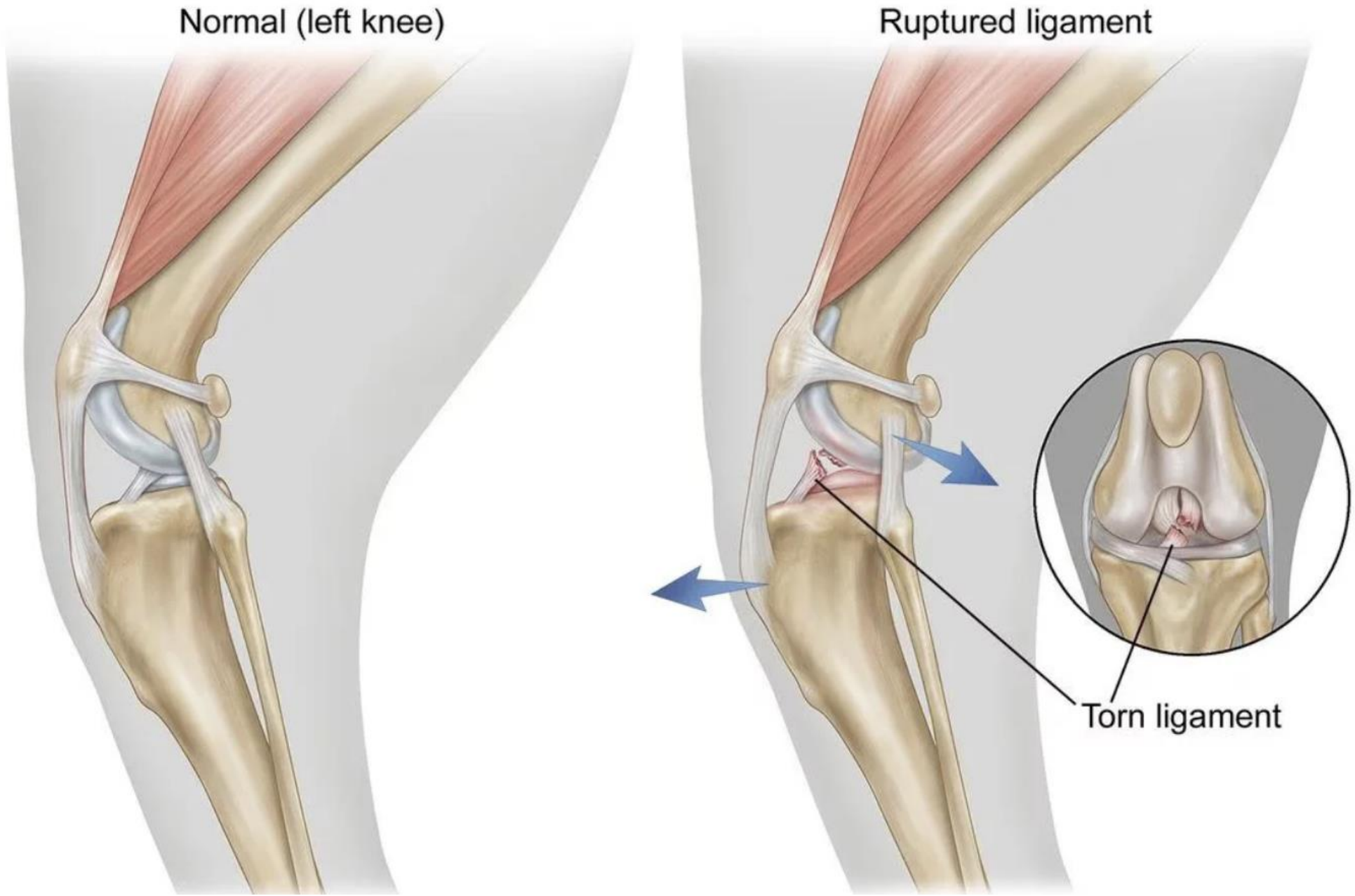


- **第三級**：一般情況下，膝蓋骨在滑車溝外，有外力狀況下，可以被推回滑車溝，外力消失，滑回異常位置



- **第四級**：一般情況下，膝蓋骨在滑車溝外，有外力狀況下，不行被推回滑車溝

Drawer test
&
Tibial compression


Normal (left knee)
Ruptured ligament
Torn ligament

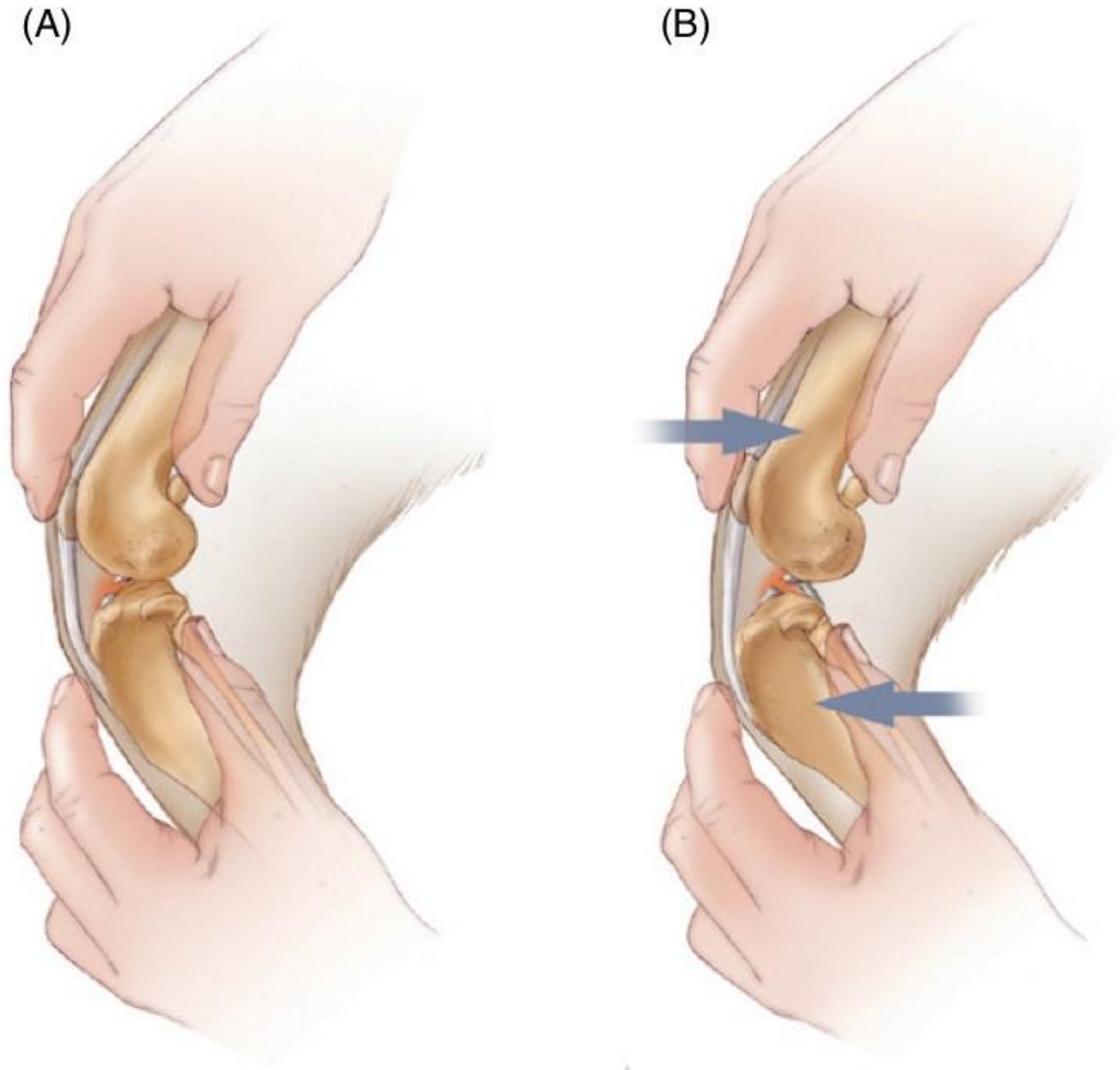


Figure 19.7 The cranial drawer test is performed by (A) placing the index finger on the patella and the thumb on the lateral fabella, with the opposite hand, the proximal tibia is grasped with the thumb placed caudal to the fibular head and the tip of the index finger on the tibial tuberosity. The lower hand is then used to (B) gently push the proximal tibia cranially, which indicates CCLD deficiency.

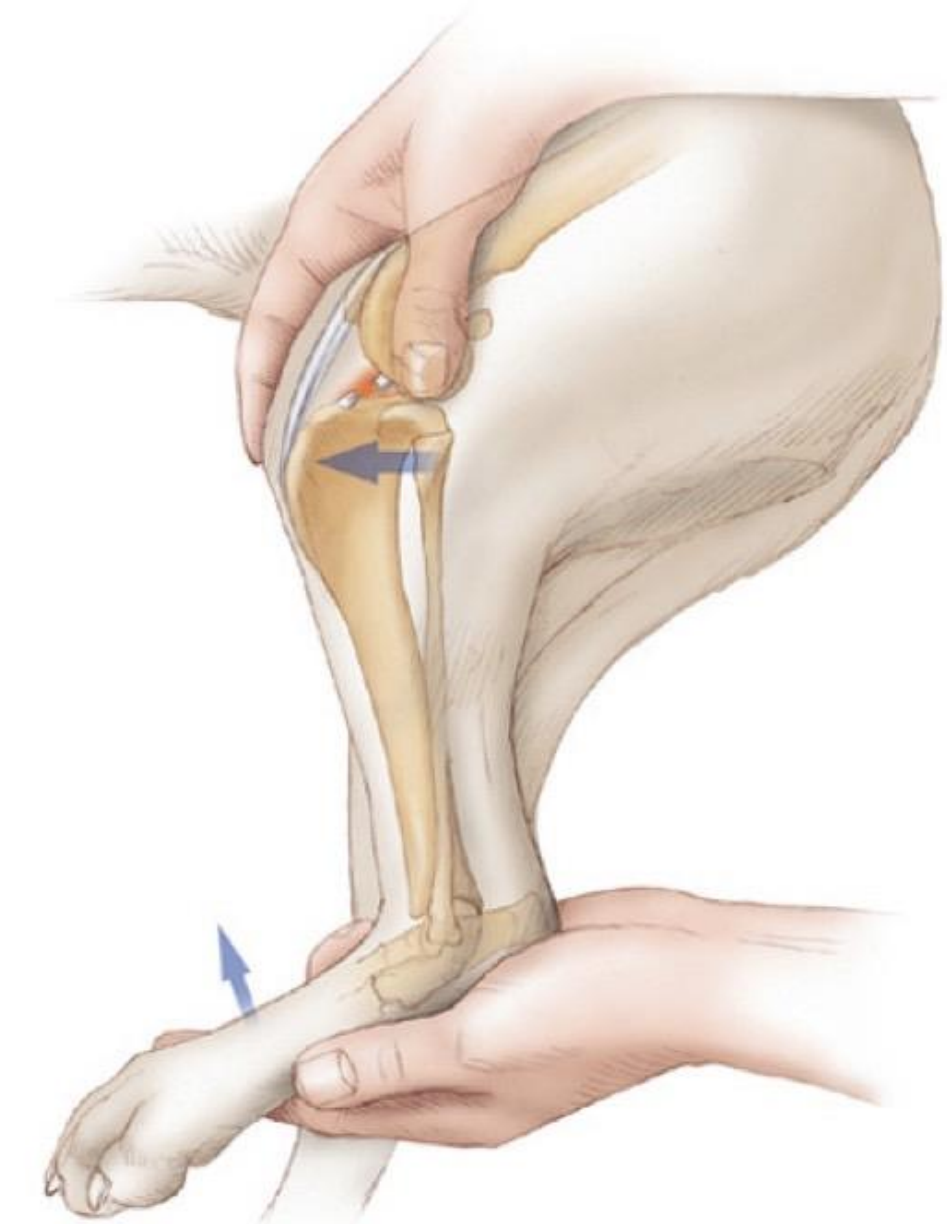
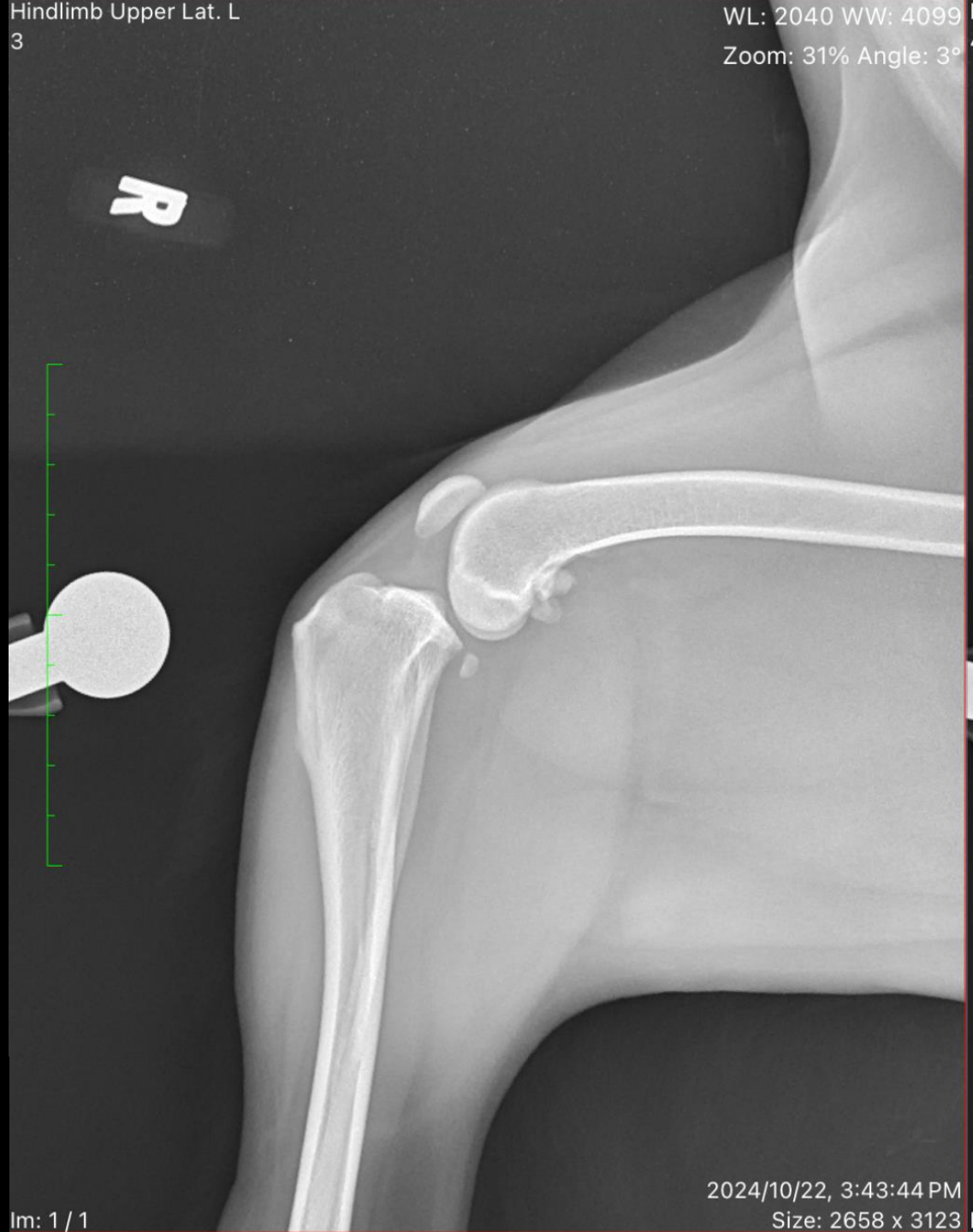
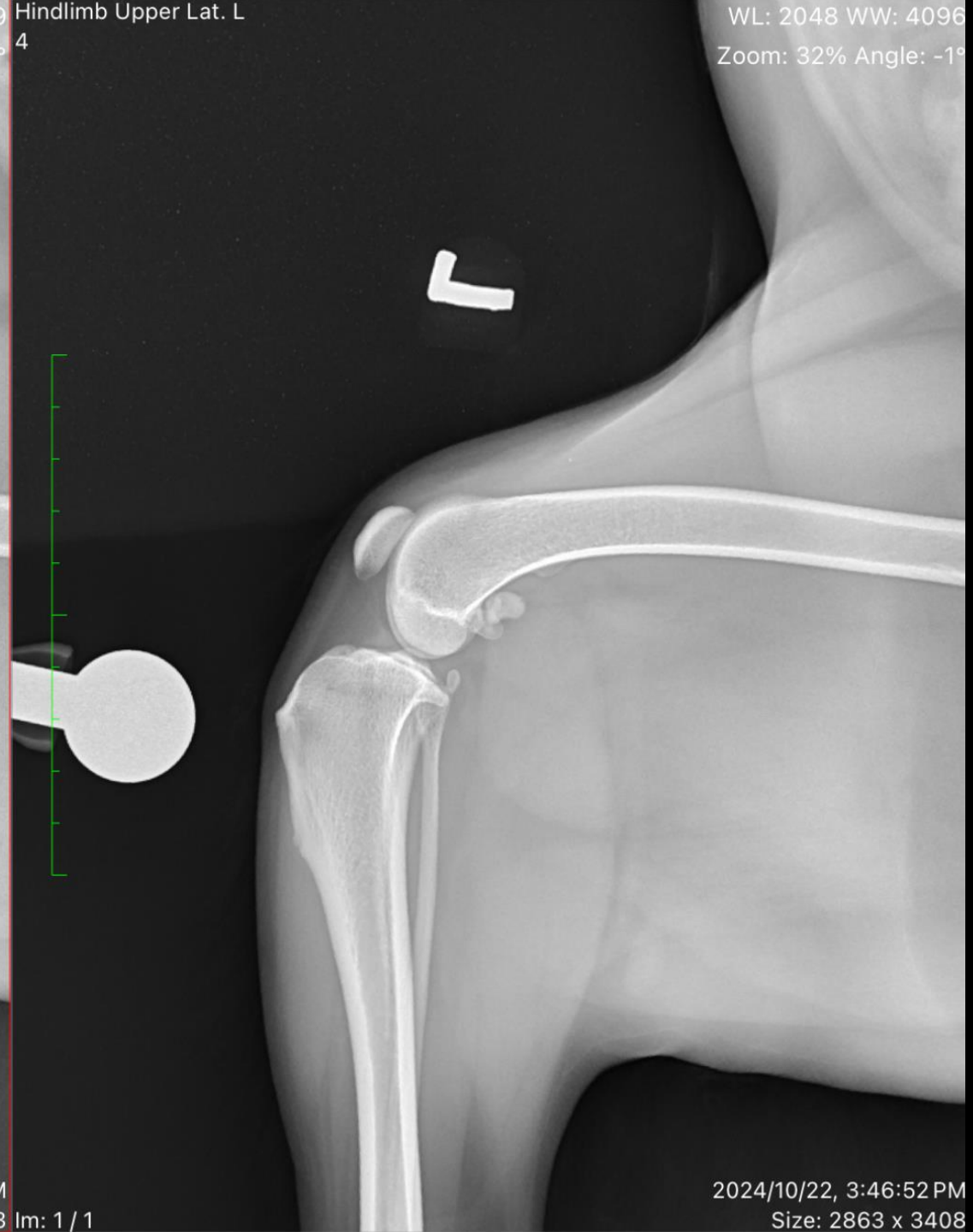


Figure 19.8 The cranial tibial compression test is performed by placing one hand over the cranial distal thigh with the index finger placed on the tibial tuberosity. The other hand is used to flex the tarsus, while the upper hand maintains the knee in extension (i.e. not allowing any flexion of the joint). Cranial tibial movement indicates CCLD deficiency and may be observed or detected with the index finger on the tibial tuberosity.




Hindlimb Upper Lat. L
3
WL: 2040 WW: 4099
Zoom: 31% Angle: 3°
R
2024/10/22, 3:43:44 PM
Size: 2658 x 3123
Im: 1 / 1


Hindlimb Upper Lat. L
4
WL: 2048 WW: 4096
Zoom: 32% Angle: -1°
L
2024/10/22, 3:46:52 PM
Size: 2863 x 3408
Im: 1 / 1

Varus / Valgus

Valgus(外翻) & Varus(內翻)

- 測試內/外側副韌帶(Medial/Lateral collateral ligament)的完整性
- 關節維持在伸展角度
- 進行內翻/外翻

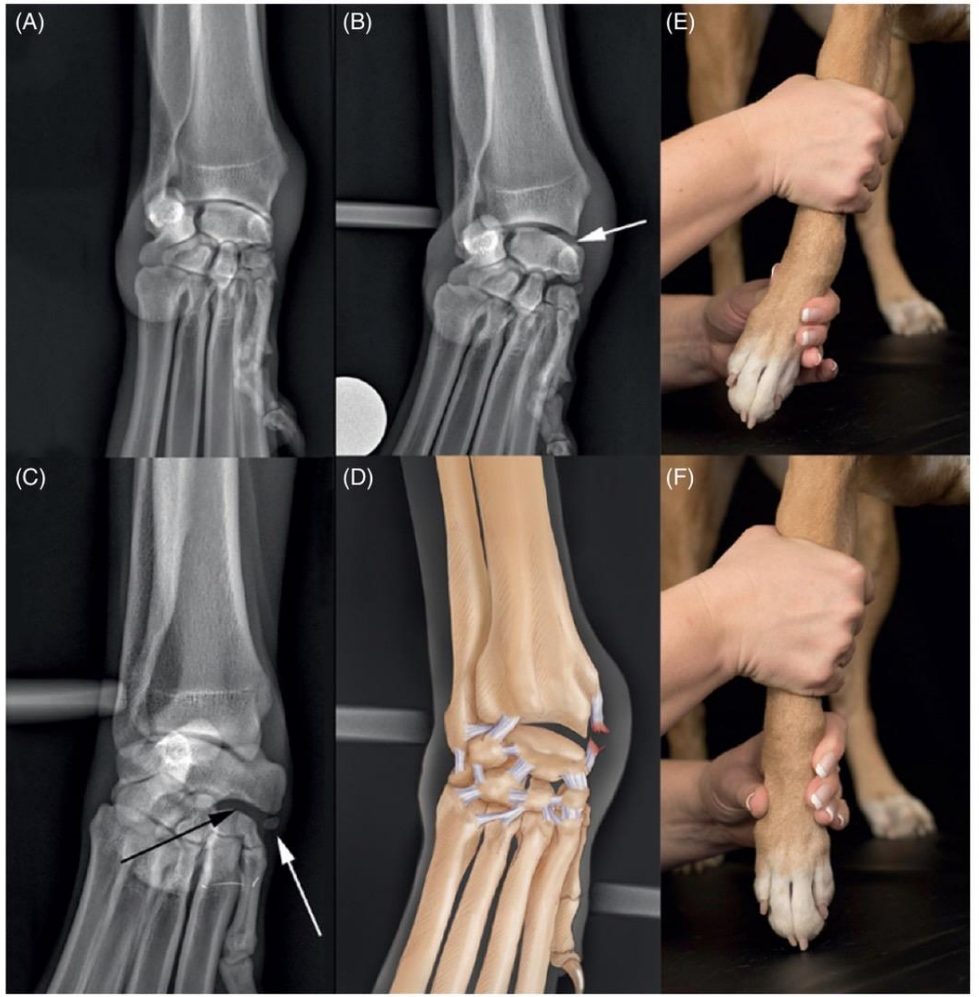


Figure 13.8 Traumatic medial collateral ligament injury in a 4-year-old GSD: (A–C) radiographs, (D) illustration of the injury, and (E, F) physical examination to detect collateral ligament injury by applying medial/later stress to these ligaments. (A) Radiographs of the affected limb showing soft tissue swelling on the medial aspect but no evidence of instability. (B) Stress radiograph of the affected limb showing an increased antebrachiocarpal joint space (white arrow) indicating disruption of the medial collateral ligament. (C) Stress radiograph of the unaffected, normal limb highlighting a normal antebrachiocarpal joint width. Note: while this is a similar stress view of the opposite limb, due to slightly different positioning, the sesamoid bone (arrow) within the abductor pollicis longus is more obvious due to lack of superimposition with other carpal bones. Similarly, the different positioning gives the appearance of a wider middle carpal joint (black arrow). (D) Illustration of image (B) depicting the torn collateral ligament and widened joint space. (E) Application of valgus stress and (F) varus stress to the carpus, testing the integrity of the medial ligament and lateral collateral ligament, respectively.

Biceps tendon

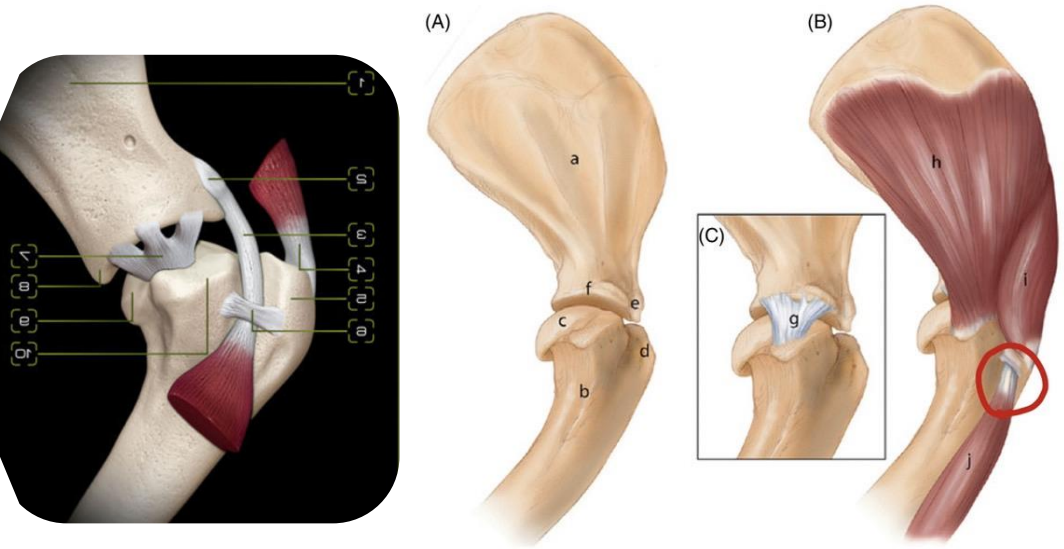


Figure 15.3 Normal anatomy of the medial shoulder for (A) osseous structures; (B) muscles; and (C) the collateral ligament of the medial shoulder: (a) scapula; (b) humerus; (c) humeral head; (d) greater tubercle; (e) supraglenoid tubercle; (f) glenoid; (g) medial glenohumeral ligament; (h) subscapularis; (i) supraspinatus; and (j) biceps brachii.

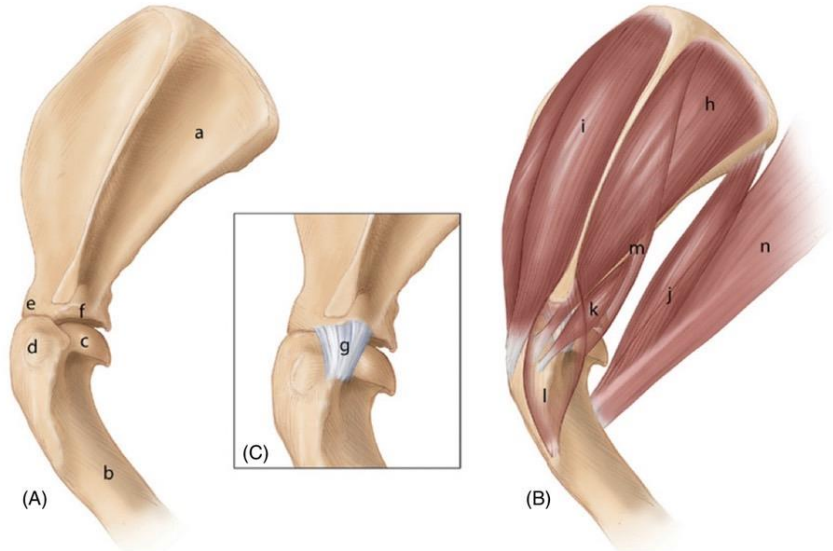


Figure 15.4 Normal anatomy of the lateral shoulder for (A) osseous structures; (B) muscles; and (C) the collateral ligament of the lateral shoulder: (a) scapula; (b) humerus; (c) humeral head; (d) greater tubercle; (e) supraglenoid tubercle; (f) glenoid; (g) lateral glenohumeral ligament; (h) infraspinatus; (i) supraspinatus; (j) teres major; (k) teres minor; (l) acromial part and (m) scapular part of the deltoideus; and (n) latissimus dorsi.

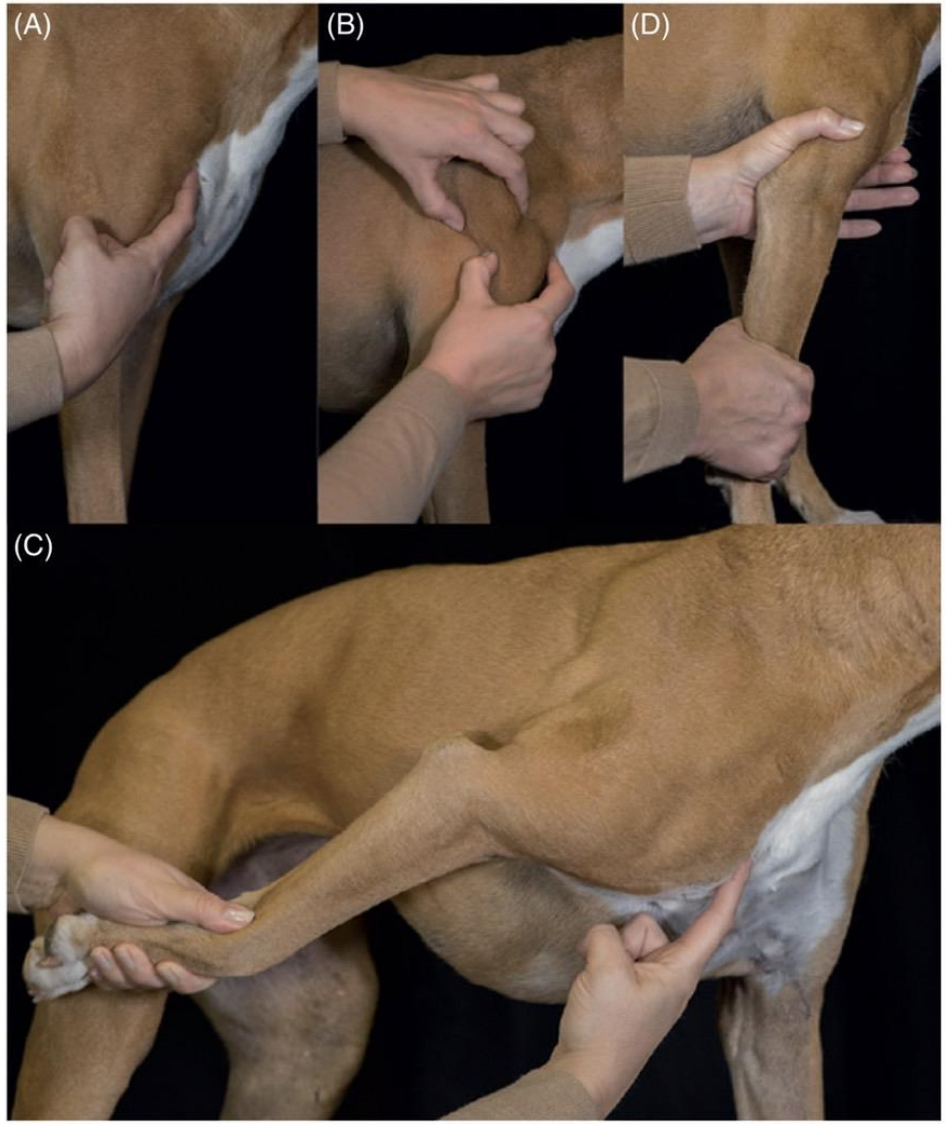
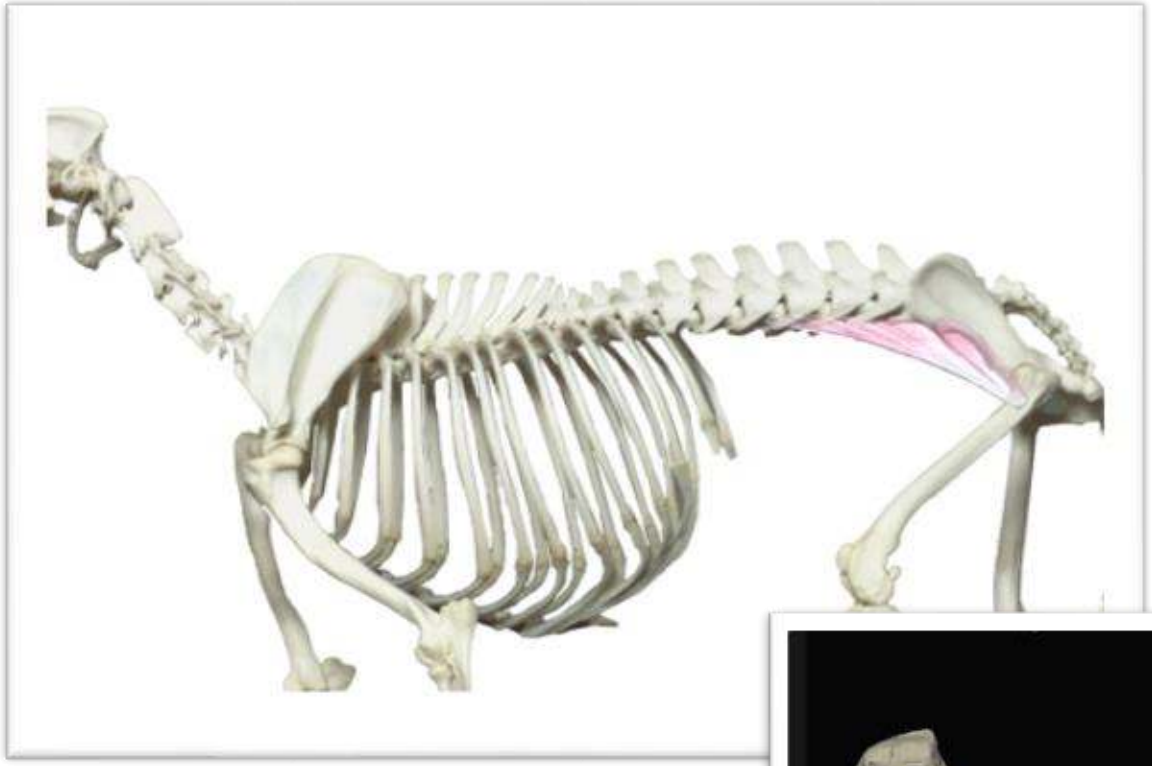


Figure 15.11 Biceps brachii palpation: (A) the biceps tendon can be digitally palpated, just medially to the greater tubercle; (B) shoulder drawer test is performed by stabilizing the scapula with one hand, while the other hand attempts to displace the humeral head cranially, relative to the supraglenoid tubercle; (C) biceps disease can frequently be detected by eliciting a painful response when placing the shoulder in flexion and hyperextending the elbow, while applying digital pressure to the biceps tendon, the so-called biceps test; (D) a painful response during the biceps test may be due to elbow disease since the elbow is hyperextended. This can easily be differentiated by performing isolated elbow hyperextension.

Iliopsoas

髂腰肌 (Iliopsoas)

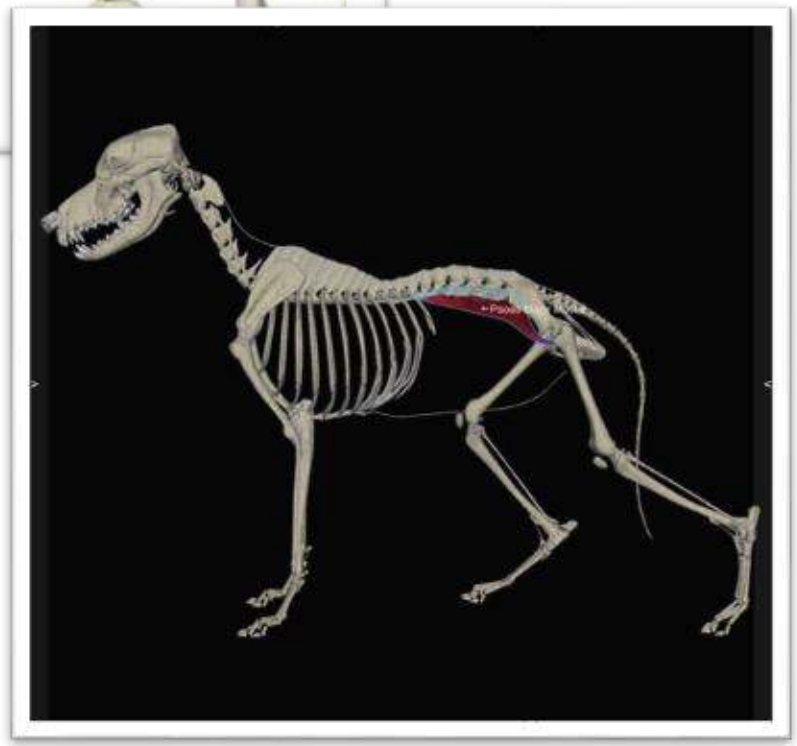


- 彎曲髖關節
- 外旋髖關節
- 彎曲腰椎

通常為繼發性/代償性疼痛

⬇

要去找原發問題



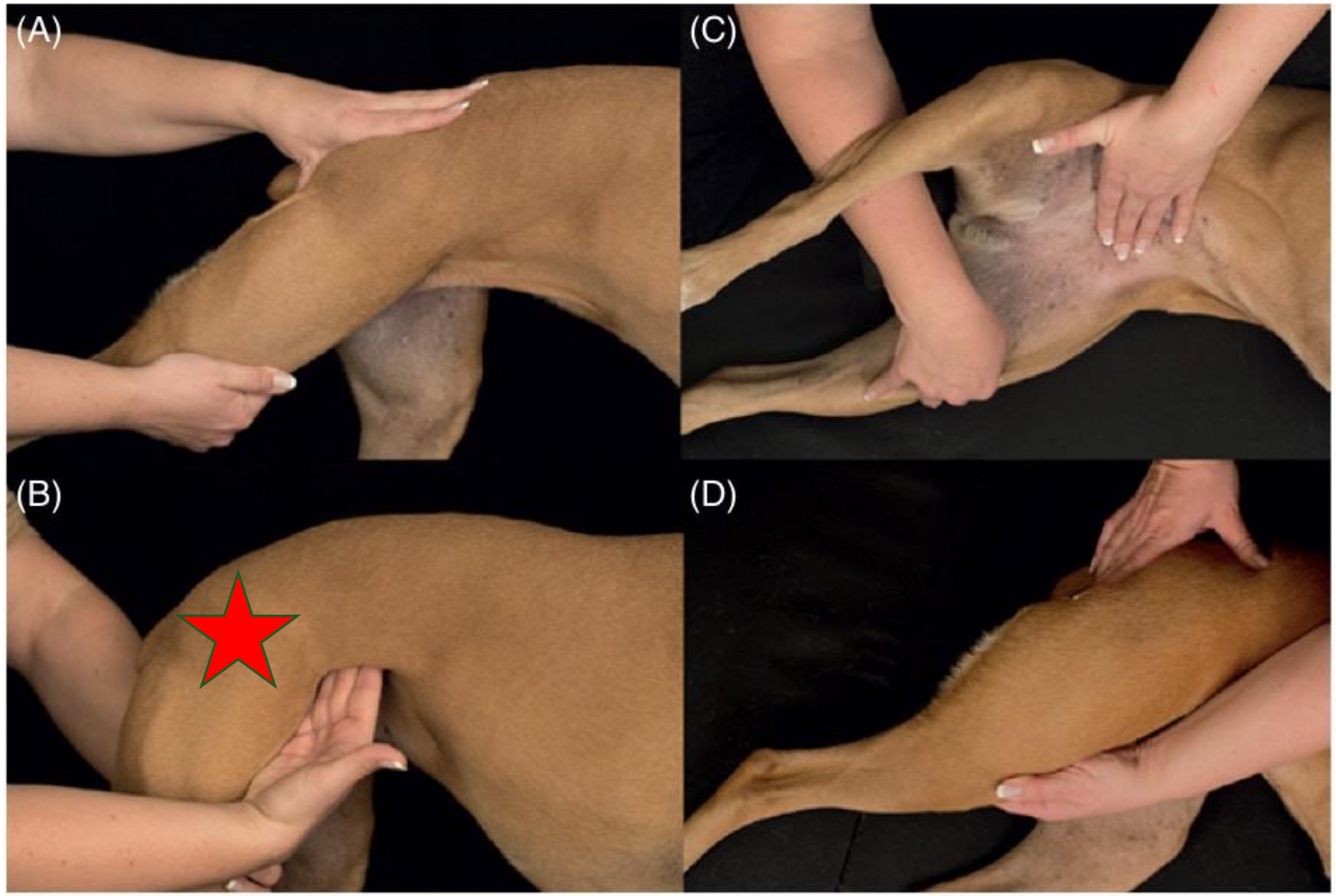


Figure 20.15 Iliopsoas palpation: the Iliopsoas muscle can be palpated (A, B) in standing or (C, D) recumbent position. Pain during (A) hip extension may be caused by the iliopsoas muscle or stifle, hip or neurologic pathology (Figure 20.8). As such, further palpation of the muscle is required to evaluate whether the source of pain is originating from the iliopsoas muscle: (B) palpation of the abdominal portion of the muscle below the vertebral column can be performed while the animal is standing or (C) while in lateral recumbency. Stretching of the muscle is performed by (D) hyperextending the spine, while simultaneously extending the hip joint and internally rotating the limb.

伸展 Hip + 內轉

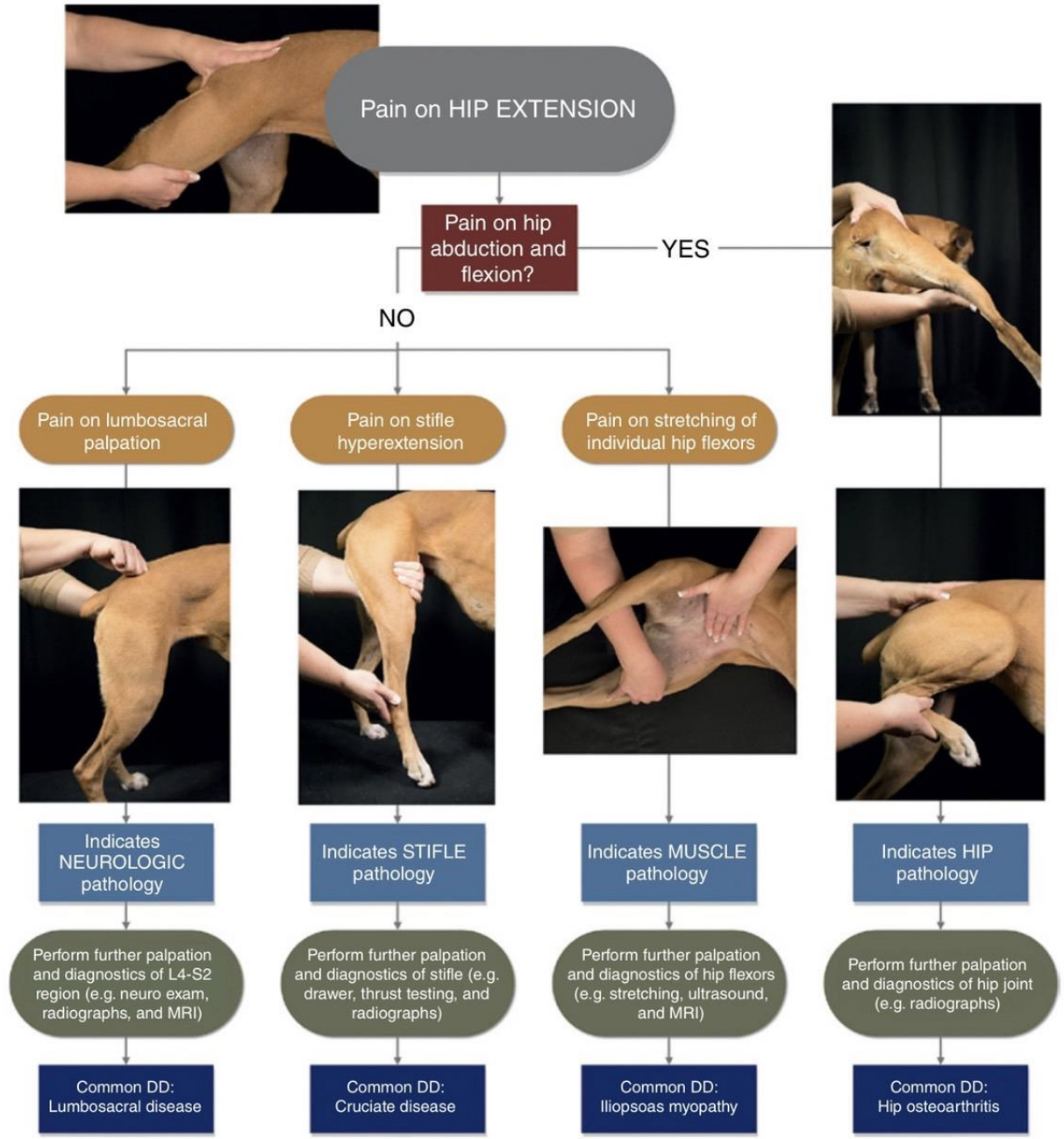


Figure 20.8 Algorithm to differentiate the four differential diagnoses for pain on hip extension: hip, stifle, neurologic, or muscular pathology.

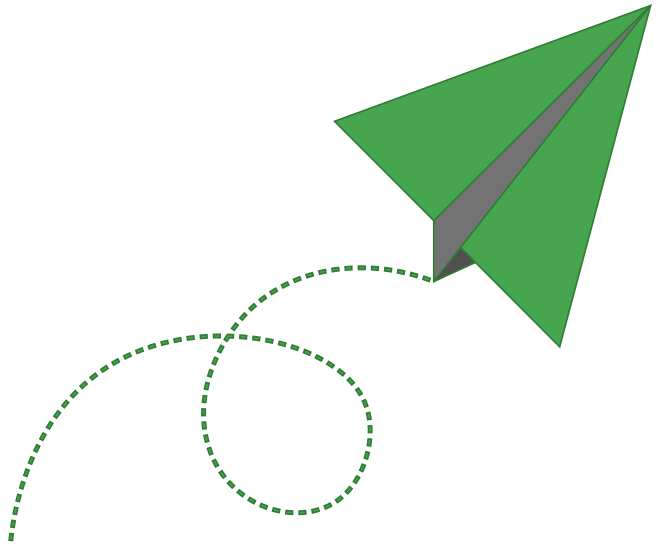
神經學檢查



檢查步驟

➢ 觀察

1. 意識、行為 (Mental status and behavior)
2. 靜止姿態 (Posture)
3. 步態 (Gait)



➢ 動手檢查

1. 姿勢反應 (Postural reactions)
2. 脊髓節段、反射檢查
3. 感覺評估（皮肌反射、觸診、痛覺）

意識、行為

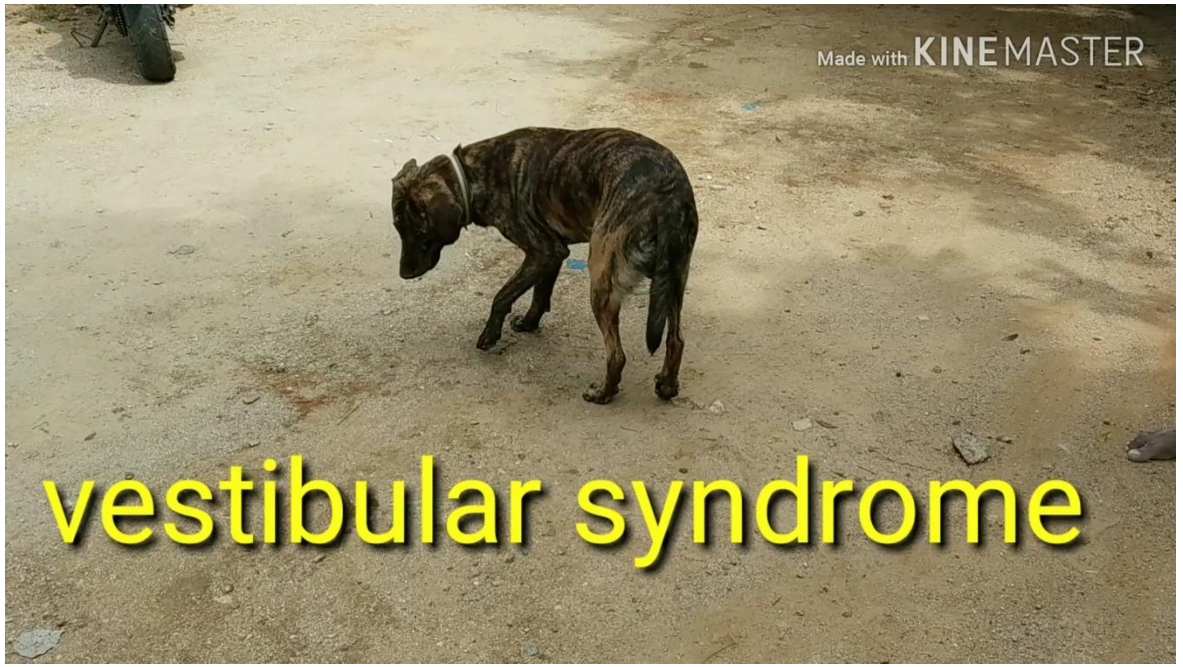
Depressed, Stuporous, Comatose

- **沉鬱 (Depressed)**：清醒狀態，但嗜睡、注意力不集中、環境刺激反應弱
- **木僵 (Stuporous)**：意識不清的狀態、環境刺激反應減弱、可藉由刺激痛覺喚醒
- **昏迷 (comatose)**：意識不清的狀態、環境刺激反應喪失(包含疼痛)
  - 兩大腦半球之瀰漫性病兆或多發性病兆、腦幹內ARAS的局部性病兆

Walking in Circles

- **大圓圈**：前腦 (Forebrain) - Ex: 右側皮質+丘腦=右轉
- **小圓圈**：前庭 (Vestibular problem) - 合併「頭部傾斜」「眼球震顫」







静止姿態

Wide/Narrow-based stance

寬站姿(Wide-based stance)	窄站姿(Narrow-based stance)
本體感覺異常	下運動神經元(LMN)
前庭	周邊神經
小腦	神經肌肉
脊髓	肌肉骨骼





Head tilt/turn

頭部傾斜(Head tilt)

✓前庭問題(Vestibular disorder)



頭部轉向(Head turn)

✓前腦問題(Cerebral disorder)



脊柱彎曲
Spinal curvature

脊椎凸起(Kyphosis)

✓胸腰椎疼痛問題 (Ex: IVDD)



脊椎側彎(Scoliosis)



脊椎凹陷(Lordosis)

✓脊椎發育異常 (Ex: hemivertebrae)



步態

Ataxia vs. Paresis

➢ **共濟失調 (Ataxia)** :

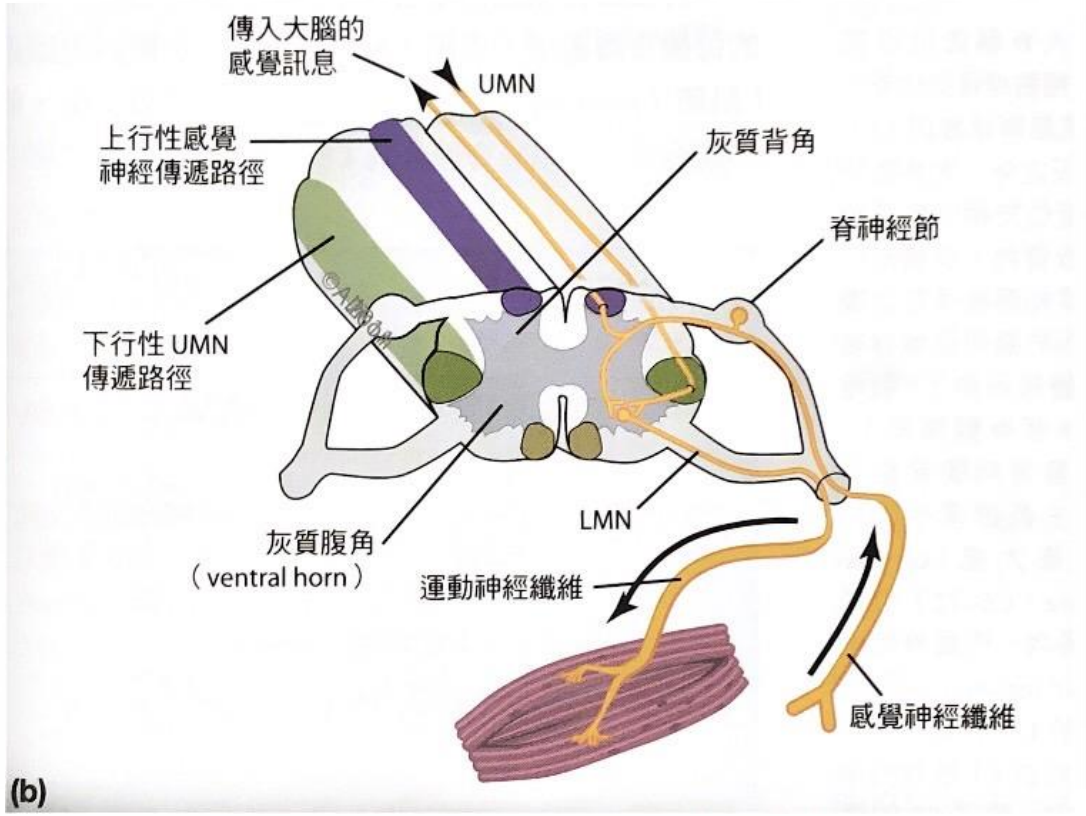
✓定義：不協調的步態

1. Sensory dysfunction
2. Proprioceptive ataxia
3. 本體感覺性、小腦性、前庭性

➢ **輕癱/全癱 (Paresis/Paralysis)** :

✓定義：自主運動能力減弱或喪失

1. Motor dysfunction
2. Lack strength, short stride, weak muscle tone
3. 腦幹、脊髓、周邊神經、肌肉



Ataxia vs. Paresis

Ataxia without paresis : 小腦 lesion

Ataxia vs. Paresis

Paresis without Ataxia : 神經肌肉 lesion

姿勢反應

姿勢反應

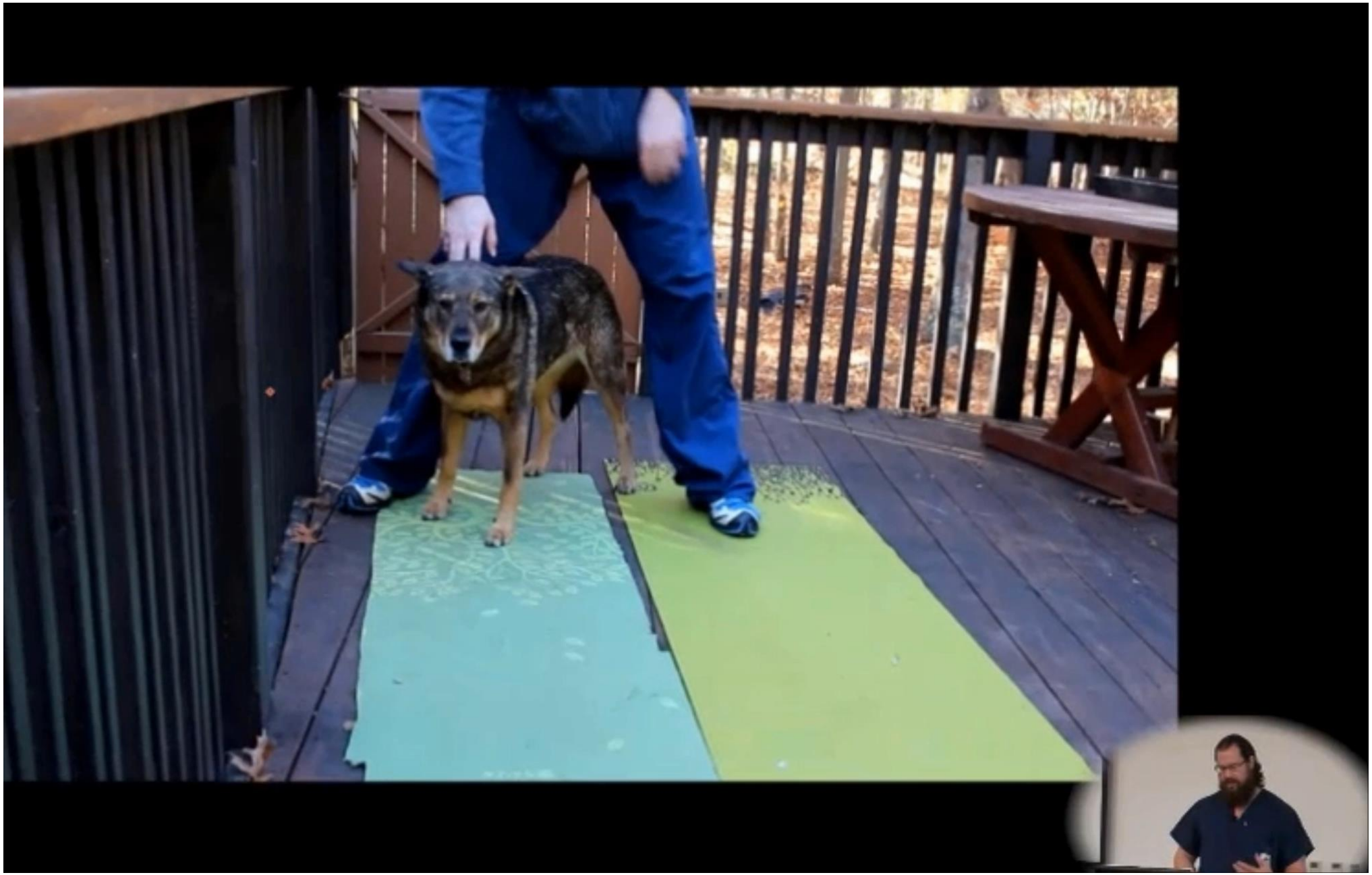
- 這些反射具有複雜的反射機制，但通常涉及：

  本體感覺受體→感覺神經元→大腦→運動神經元→骨骼肌

- 整套完好的神經系統才能完成，判斷是否為神經系統問題的良好指標!!
- 放置反應 (Proprioceptive placing)
- 跳躍反應 (Hopping)

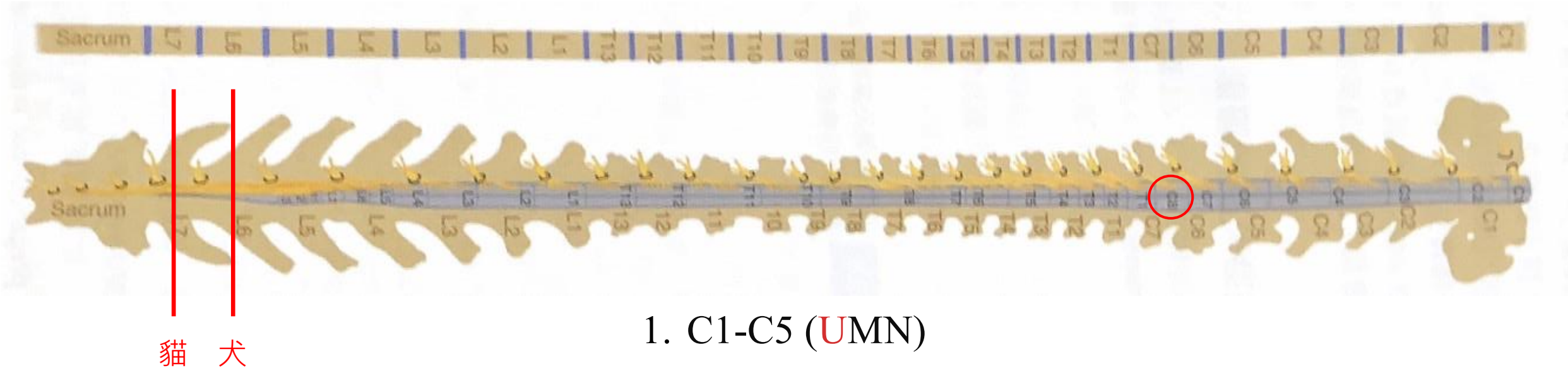






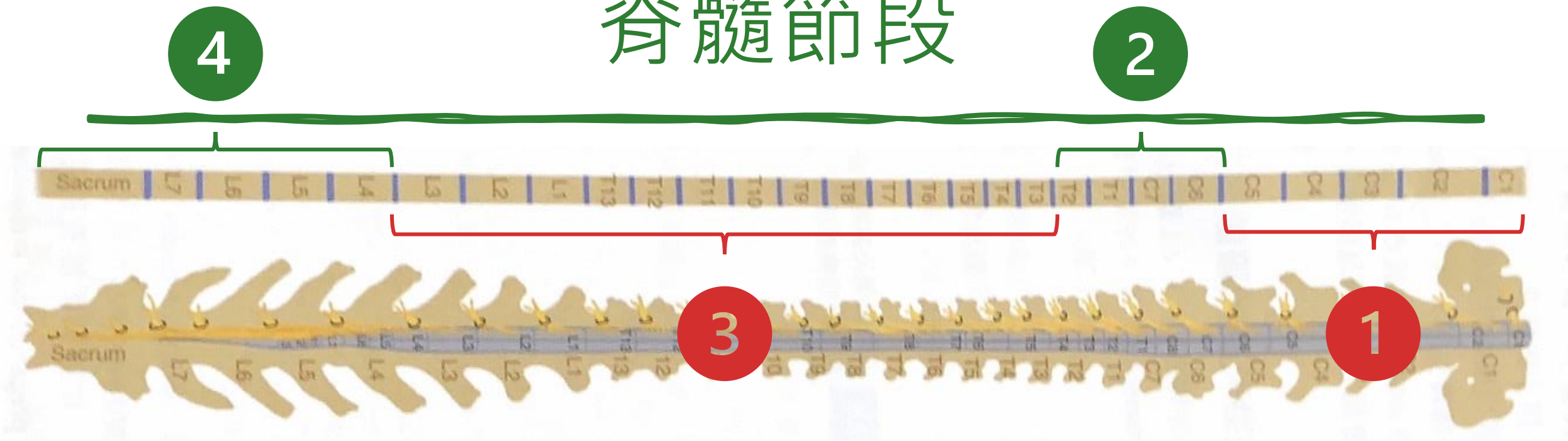
脊髓節段、反射檢查

脊髄節段



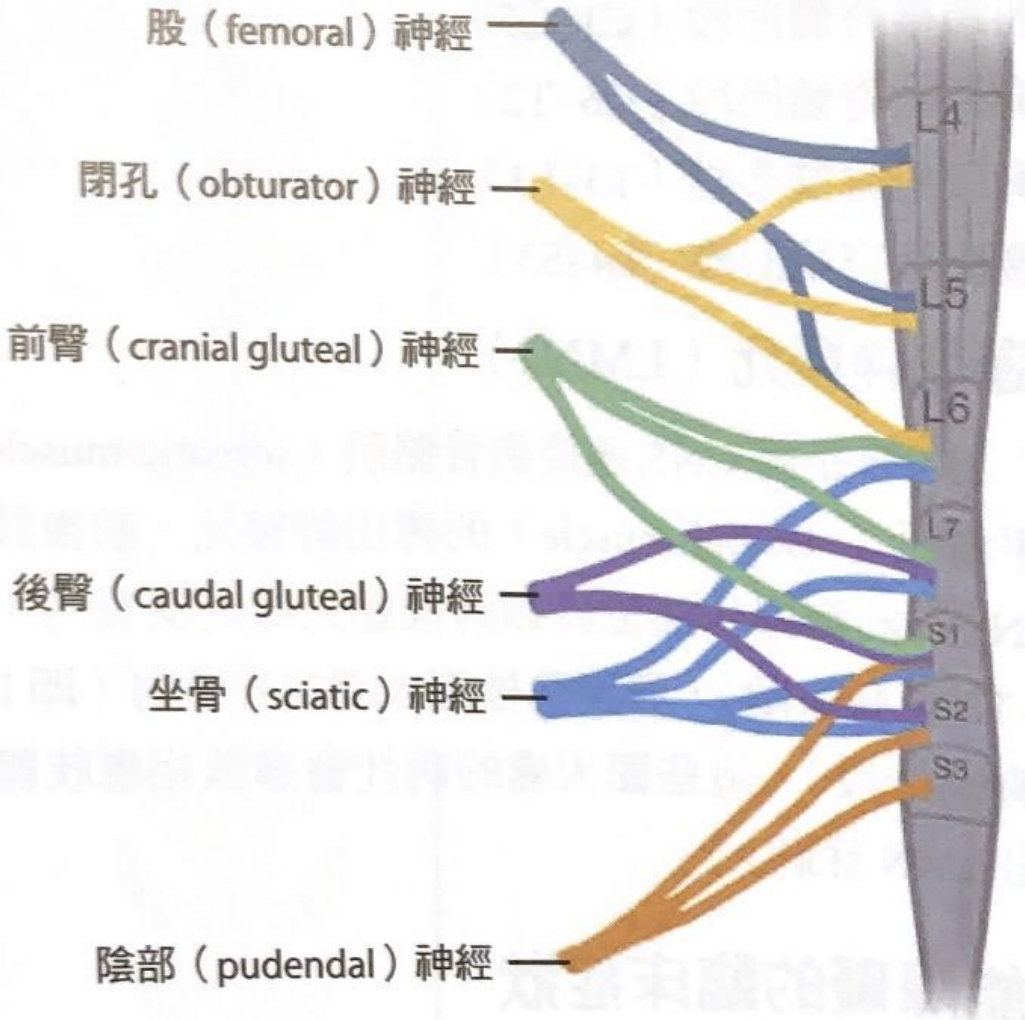
1. C1-C5 (UMN)
2. C6-T2 (LMN)
3. T3-L3 (UMN)
4. L4-S3 (LMN)

脊髓節段

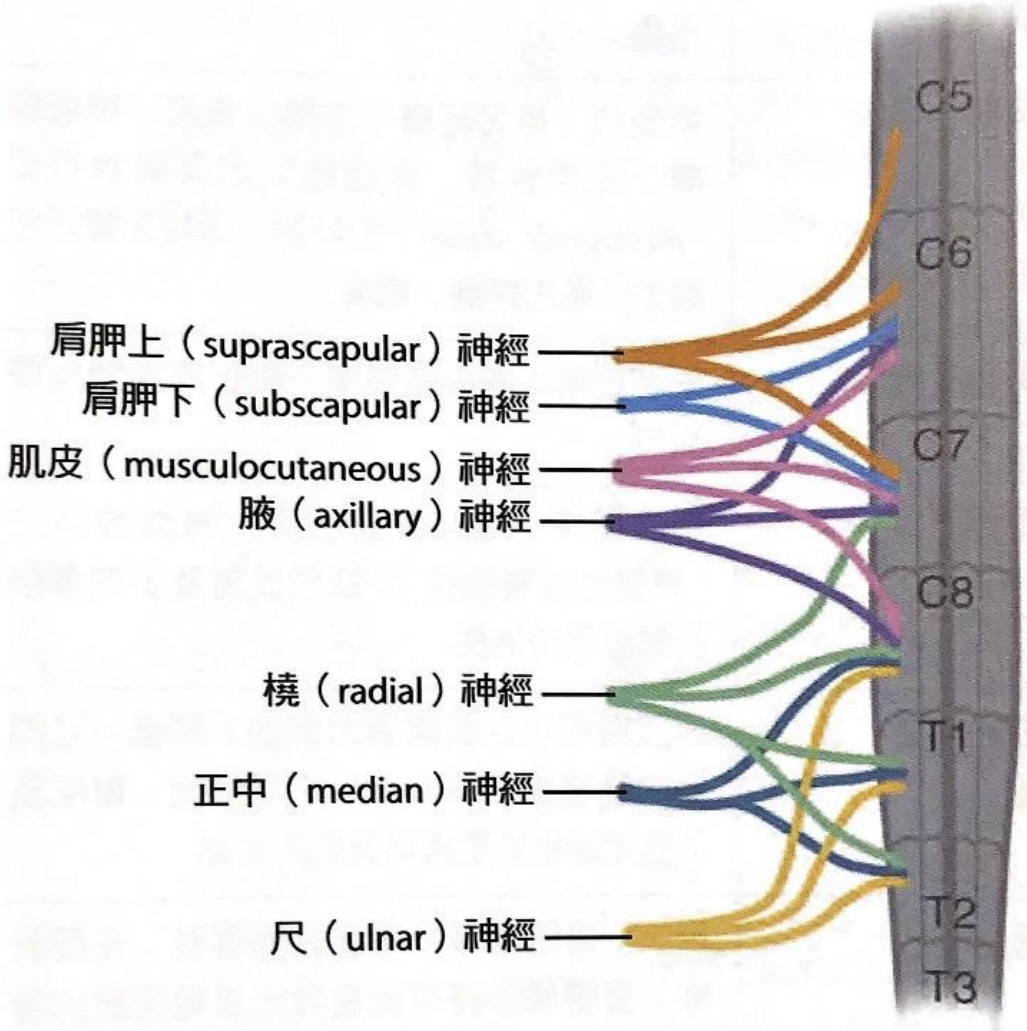


1. C1-C5 (UMN)
2. C6-T2 (LMN)
3. T3-L3 (UMN)
4. L4-S3 (LMN)

4


股（femoral）神經
閉孔（obturator）神經
前臀（cranial gluteal）神經
後臀（caudal gluteal）神經
坐骨（sciatic）神經
陰部（pudendal）神經
L4
L5
L6
L7
S1
S2
S3

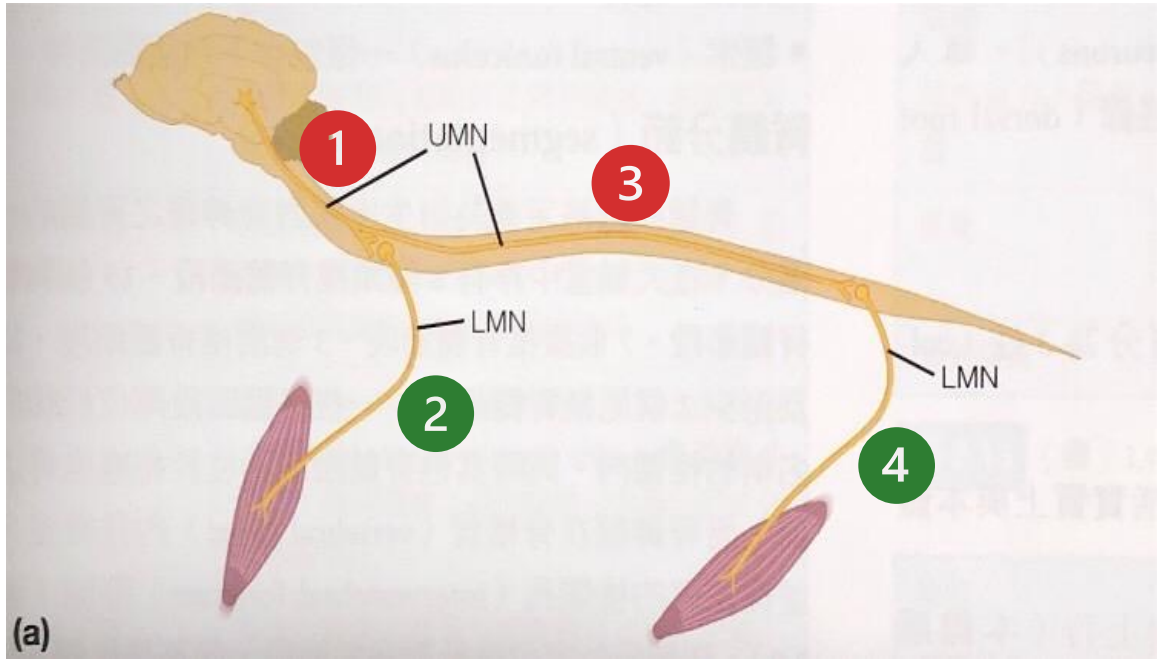
2


肩胛上（suprascapular）神經
肩胛下（subscapular）神經
肌皮（musculocutaneous）神經
腋（axillary）神經
橈（radial）神經
正中（median）神經
尺（ulnar）神經
C5
C6
C7
C8
T1
T2
T3

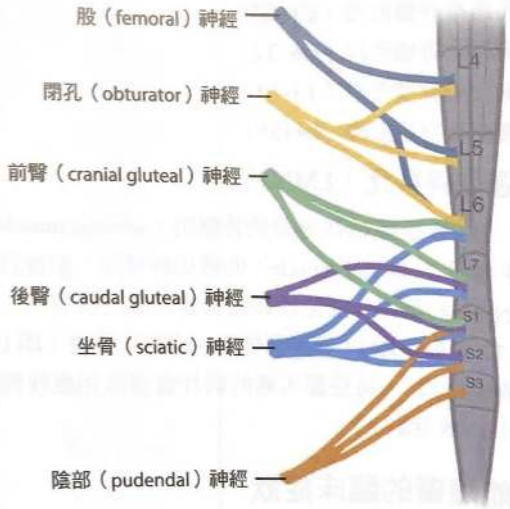
UMN vs. LMN

- **UMN (Upper Motor Neuron) :**
  - 源自大腦的傳出神經元，間接與LMN進行突觸接合
  - UMN對肌肉伸展反射 (Myotatic reflexes)具有抑制作用

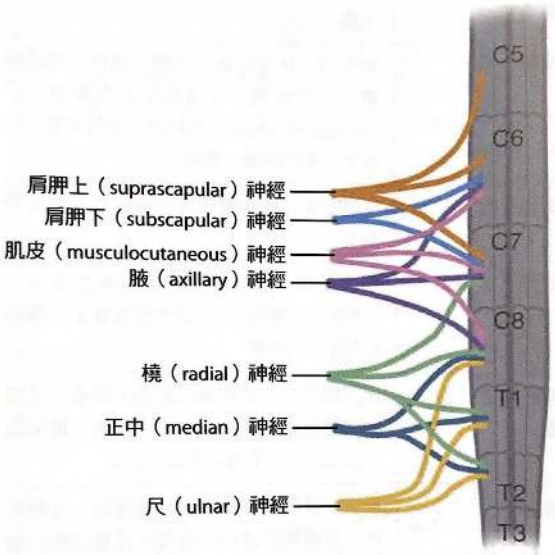
- **LMN (Lower Motor Neuron) :**
  - 連接中樞神經系統與作用器官的肌肉
  - LMN為神經鏈 (Chain of neurons)中的最後一個神經元
  - 脊髓反射 – Patella reflex, Withdrawal…



脊髓反射(LMN)



	脊髓反射	脊髓节段	神经
前肢	屈肌回缩反射 Withdrawal	C6-T2	腋神经 Axillary n. 肌皮神经 Musculocutaneous n. 正中神经 Median n. 尺神经 Ulnar n.
后肢	膝反射 Patellar	L4-6	股神经 Femoral n.
	屈肌回缩反射	L6-S2	坐骨神经 Sciatic n.
肛门	肛门反射	S1-Cd5	阴部神经 Pudendal n. 尾神经 Caudal nerves n.
膀胱	触诊 - 括约肌张力 - 膀胱张力	S1-3	骨盆神经 Pelvic n. 阴部神经 Pudendal n.



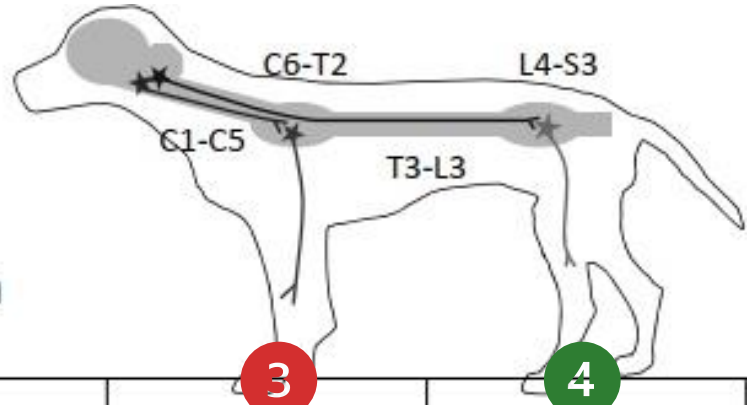
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Withdrawal test



Patellar reflex test

UMN sign vs. LMN sign



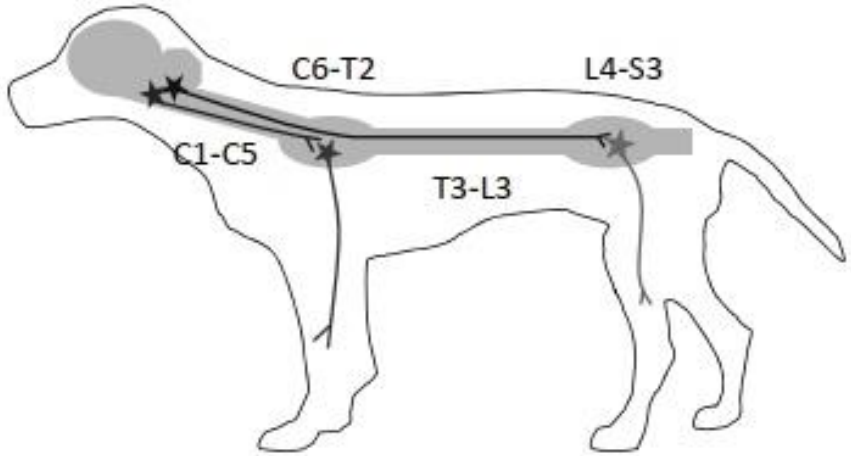
UMN vs. LMN lesion

1	2	3	4
C1-C5 脊髓节段	C6-T2 脊髓节段	T3-L3 脊髓节段	L4-S3 脊髓节段
步态、运动功能、姿态反应可能呈现… 經過大腦			
四肢异常	四肢异常	前肢正常 后肢异常	前肢正常 后肢异常
脊髓反射 LMN 不需經過大腦			
四肢正常	前肢下降 后肢正常或↑	四肢正常	前肢正常 后肢下降

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	1	2	3	4
	C1-C5	C6-T2	T3-L3	L4-S3
前肢	UMN	LMN	normal	normal
後肢	UMN	UMN	?	LMN

脊髓反射與肌肉張力



	上运动神经元病灶 UMN lesion	下运动神经元病灶 LMN lesion
肌肉萎缩	较晚发生、轻微	早发生、严重
脊髓反射	正常 - ↑	↓ - 消失
肌肉张力	正常 - ↑	↓

From 張雅珮 老師 2019 CCRV session1

UMN對肌肉伸展反射(Myotatic reflexes)具有抑制作用

UMN gait vs. LMN gait

- **UMN lesion**
  - Stiff 僵硬
  - Spastic 痙攣
  - Long-strides
- **LMN lesion**
  - Looks lame
  - Support problem
  - Short-strides

	脊髓反射	脊髓节段	神经
前肢	屈肌回缩反射 Withdrawal	C6-T2	腋神经 Axillary n. 肌皮神经 Musculocutaneous n. 正中神经 Median n. 尺神经 Ulnar n.
后肢	膝反射 Patellar	L4-6	股神经 Femoral n.
	屈肌回缩反射	L6-S2	坐骨神经 Sciatic n.
肛门	肛门反射	S1-Cd5	阴部神经 Pudendal n. 尾神经 Caudal nerves n.
膀胱	触诊 - 括约肌张力 - 膀胱张力	S1-3	骨盆神经 Pelvic n. 阴部神经 Pudendal n.

膀胱

	上运动神经元病灶 UMN bladder	下运动神经元病灶 LMN bladder
病灶	位于S1-3之前	S1-3 骨盆神经 Pelvic n. 阴部神经 Pudendal n.
临床症状	尿液滞留 Urinary retention	尿失禁 Urinary incontinence
触诊	膀胱壁张力佳 膀胱大且紧绷	膀胱壁张力差 膀胱大但松垮
挤尿	困难	容易 (但不容易完全排空)

From 張雅珮 老師 2019 CCRV session1

交叉性伸肌反射
Crossed-extensor reflex

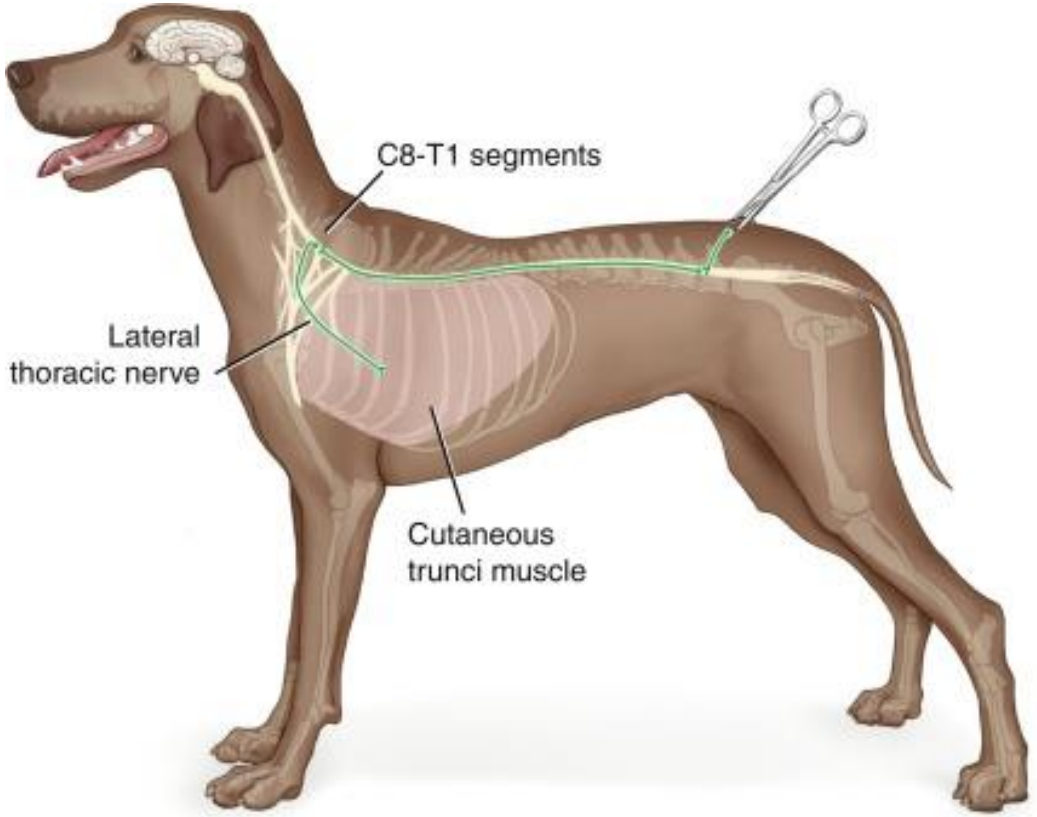
- 測試屈肌回縮反射時對側腳會呈現伸展反應 ➡ UMN Sign!!!
- 前肢出現Crossed-extensor reflex ➡ C1-C5
- 後肢出現Crossed-extensor reflex ➡ C1-L3



皮肌反射、觸診、痛覺

皮肌反射
Cutaneous trunci reflex

- 藉由夾捏 T2-L5之間的背側軀幹皮膚來誘發反射
- 感覺神經-脊髓-位於C8~T1的外側胸神經 (Lateral thoracic nerve)-運動神經-皮肌收縮
- 從髂骨翼(Ilial wings)開始，脊柱兩側由後往前夾; 有反應時往前數2節(0-4節)
- 近一步縮小T3-L3之脊髓病兆



觸診

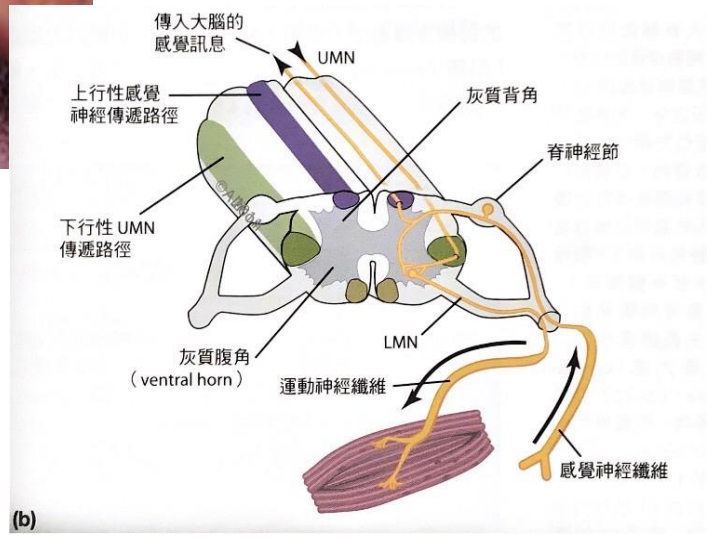
- **頭部**：不對稱、疼痛、囪門未閉合
- **脊椎**：沿著橫突方向，略施向下的壓力於棘突上
- **肢體**：神經性萎縮、骨骼肌肉問題



痛覺
Pain sensation

- 需要經過大腦的行為反射
  (Ex:咬你、轉頭看你)
- 感覺神經位於脊髓白質深處-嚴重脊髓損傷才會傷到
- 若有自主運動就不用測試此項





胸腰椎IVDD癱瘓分級



胸腰椎椎間盤疾病分級

www.wondervet.com.tw

神經學分級	行走	自主排便排尿	其他症狀	手術成功率	手術後恢復時間	*脊髓軟化發生機率
1	V	V	背痛	>95%	<2周	0%
2	V	V	步態不穩	95%	<2周	0%
3	X	V	肢體仍有運動能力	93%	<2周	0.6%
4	X	X	有深層痛覺	95%	1-4周	2.7%
5	X	X	無深層痛覺	64%	5-10周	14.6%

*脊髓軟化為胸腰椎椎間盤疾病最嚴重的併發症